Cellpose-Based Segmentation for Extracting Morphological Features from Drug-Treated Cancer Cell Images

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1 Introduction

High-content imaging (high-throughput microscopy) is widely used to study how biological samples respond to chemical exposure. For example, some drug compounds may change morphological characteristics in cells by affecting cytoskeleton, DNA replication, or protein regulation. These changes create a measurable phenotypic fingerprint: cells may change their shape, size, nuclear morphology, actin organisation, or microtubule structure depending on the compound's mechanism of action.

However, manual evaluation of microscopy images misses many details required for comparison. Therefore, computational phenotypic profiling is used. It converts microscopy images into numerical features that can capture even subtle cellular changes and enable classification of drug mechanisms of action (Ljosa et al., 2013).

This project uses an interpretable subset of cells treated with 5 compounds came from the bigger image set of cancer cells fluorescent microscopy (Broad Bioimage Benchmark Collection).

The aim of the project is to develop a pipeline for cell segmentation, extracting morphometric features, and comparing cellular phenotypes induced by different compounds. This strategy was introduced in previous studies by (Caie et al., 2010) and (Ljosa et al., 2013).

2 Survey of the state of the art

Image-based profiling in high-content microscopy treats microscopic images not only as illustrations, but as sources of quantitative cellular phenotypic profiles. In such experiments, cells are stained across multiple channels, for example nuclei, cytoskeleton, and microtubules, and each cell is then described by measurements of size, shape, intensity, texture, and spatial organization. This approach makes it possible to compare drug-induced phenotypes and group compounds by similarity of morphological response, including the prediction of mechanism of action (Caie et al., 2010; Ljosa et al., 2013). An important step in the field was the development of open pipelines such as CellProfiler, which made large-scale segmentation and feature extraction reproducible for high-content screens, where manual image assessment becomes a major bottleneck (Carpenter et al., 2006).

Classical segmentation methods have long been based on thresholding, morphological operations, and watershed algorithms. Otsu's method automatically selects a threshold from the gray-level histogram by maximizing the separability of foreground and background classes, and is therefore often used as a simple first step for extracting an object mask (Otsu, 1979). Watershed segmentation, historically introduced as a way to separate regions by analogy with topographic catchment basins, is especially useful for separating touching objects when reliable markers for individual cells or nuclei are available (Beucher and Lantuéjoul, 1979; Vincent and Soille, 1991). However, in cell images this approach is sensitive to noise, non-uniform intensity, and dense clusters: incorrect markers can lead to over-segmentation or merging of neighboring cells. For this reason, marker-controlled watershed requires careful selection of nuclear markers, distance transform, and the mask defining the region of cell growth, as well as visual inspection of the result (Koyuncu et al., 2012).

The current state of the field is largely connected with deep learning. U-Net became one of the foundational architectures for biomedical image segmentation because it combines a contracting path for contextual information with an expanding path for precise pixel localization, while remaining applicable even with a limited number of annotated images through data augmentation (Ronneberger et al., 2015). Specialized models for cells and nuclei were later developed on this basis. StarDist performs instance segmentation by predicting star-convex polygons and is particularly suitable for dense nuclear images, where simple pixel classification can incorrectly merge adjacent objects (Schmidt et al., 2018). Cellpose moved toward generalist segmentation: the model was trained on diverse cell images and predicts vector flows that group pixels into individual cell masks. As a result, Cellpose can often be applied without training a new model and with fewer manual adjustments than other pipelines (Stringer et al., 2021).

Nevertheless, universal segmentation does not solve the entire task of morphological profiling. After masks have been obtained, it is necessary to extract single-cell features and preserve their connection to metadata: compound, concentration, well, plate, imaging channel, and segmentation parameters. New tools aim to combine segmentation and feature extraction into a more automated workflow. For example, Cellpose+ extends Cellpose by computing morphological metrics from segmented stained cells (Huaman et al., 2024), while mod-

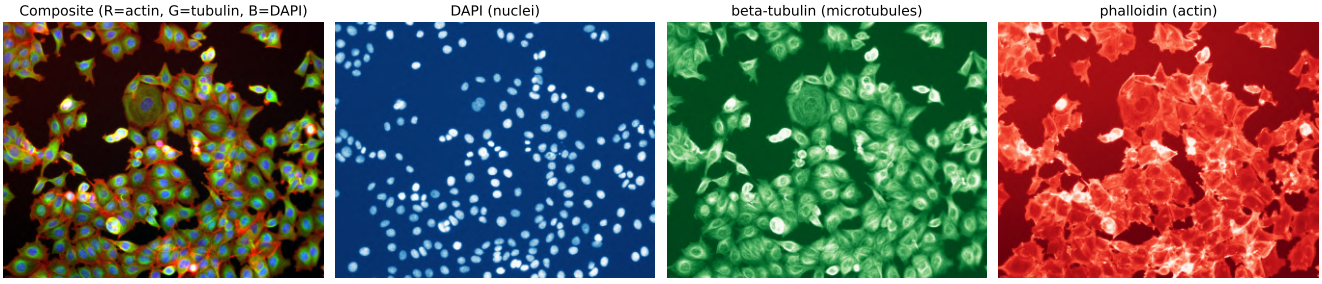


Figure 1: Dmso reference field Week9 39221 D02

ern deep learning pipelines for Cell Painting aim to reduce manual configuration and remain compatible with CellProfiler-like morphological descriptors (Zovathi and Kainz, 2026). In parallel, another direction is developing models that extract phenotypic representations directly from images, sometimes without explicit single-cell segmentation, which can be faster but less interpretable at the level of individual cells (Pawlowski et al., 2016; Ando et al., 2017; Harrison, 2023). Therefore, for this project, the most justified strategy is to compare a simple watershed baseline, StarDist-guided nuclear markers, and Cellpose-based whole-cell segmentation: this set reflects the development of the field from interpretable classical algorithms to modern generalist deep learning models, while the reliability of subsequent morphometric feature extraction strongly depends on segmentation quality.

3 Methodological contribution

3.1 Dataset description

The project uses the image set (Broad Bioimage Benchmark Collection), (Caie et al., 2010), (Ljosa et al., 2012) that is microscopy images of MCF-7 breast cancer cells treated for 24 h with a collection of 113 small molecules at eight concentrations. The cells were fixed, labeled for DNA, F-actin, and Beta-tubulin, and imaged by fluorescent microscopy (Caie et al., 2010). Metadata contains linking images to compounds, plates, wells, and concentrations, mechanism-of-action for selected compound-concentration pairs.

Table 1: Channels description

Channel	Shows	Colour
DAPI	Nucleus	Blue
beta-tubulin	Microtubules	Green
Actin	Cytoskeleton	Red

Each field of view is therefore acquired in 3 channels: the signal from nuclei, microtubules, and actin separately. Each channel is stored as a grayscale TIFF. Several cellular structures are included because a single structure does not describe the entire morphology of a cell. Nuclei help count cells and serve as markers for segmentation. Microtubules provide information about cytoskeletal organisation. The actin cytoskeleton provides information about cell boundaries and cell shape. In other words, DAPI helps show where the cells are, while tubulin and actin help reveal how drug treatment changes cellular structure.

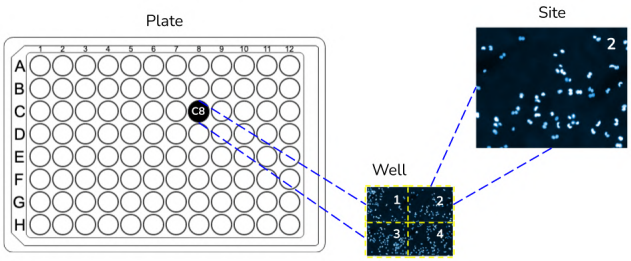


Figure 2: Plate-well-field hierarchy

DAPI (4',6-diamidino-2-phenylindole) is a fluorescent DNA-binding dye and is therefore used to visualize cell nuclei. In the image-based profiling assay used for this type of data, the actin cytoskeleton is visualized with fluorescently labeled phalloidin, while the microtubule cytoskeleton is detected using an anti-β-tubulin antibody with a fluorescent secondary antibody (Caie et al., 2010). Actin filaments and microtubules are two major components of the cytoskeleton: actin filaments contribute to cell-surface shape, protrusions, movement, attachment to the substratum, and muscle contraction, whereas microtubules support intracellular transport and reorganize into the mitotic spindle during cell division (Alberts et al., 2002).

The metadata are stored in two CSV files:

- `BBBC021_v1_image.csv` is an index of all images. For each field of view, it contains the filenames of the three channels, the plate, well, site, concentration, and compound name.
- `BBBC021_v1_moa.csv` is a lookup table mapping each compound to its mechanism of action (MOA).

From `BBBC021_v1_image.csv`, we learn that each row represents one field of view. There are 13,200 fields in total. For each field, the table contains information about which three TIFF files correspond to the same location on the plate, which compound the cells were treated with, the concentration, and the well. We use this table to get the image paths and select the required ZIP archives.

In total, the dataset contains 113 compounds x 8 concentrations x 3 replicates.

Table 2: Mechanisms of action (MoA)

	moa	compounds
0	DMSO	DMSO
1	Actin disruptors	cytochalasin B, cytochalasin D, latrunculin B
2	Aurora kinase inhibitors	AZ-A, AZ258, AZ841
3	Cholesterol-lowering	mevinolin/lovastatin, simvastatin
4	DNA damage	chlorambucil, cisplatin, etoposide, mitomycin C
5	DNA replication	camptothecin, floxuridine, methotrexate, mitoxantrone
6	Eg5 inhibitors	AZ-C, AZ138
7	Epithelial	AZ-J, AZ-U, PP-2
8	Kinase inhibitors	PD-169316, alsterpaullone, bryostatin
9	Microtubule destabilizers	colchicine, demecolcine, nocodazole, vincristine
10	Microtubule stabilizers	docetaxel, epothilone B, taxol
11	Protein degradation	ALLN, MG-132, lactacystin, proteasome inhibitor I
12	Protein synthesis	anisomycin, cyclohexamide, emetine

As shown in Figure 2, all images were acquired in 55 96-well plates, which are laboratory plates with wells where cell cultures are grown. One plate corresponds to one ZIP archive of 700 MB. Each plate has 96 wells; each well contains either the control condition (DMSO) or one treatment with one compound at one concentration. A site / field of view is one photographed area within a well, usually 4 sites per well. The three TIFF files are the three channels of the same site.

Each image is a microscopic field of view containing several dozen MCF-7 cells.

3.2 Compound selection

As shown in Table 2, it can be seen that all compounds in the dataset can be divided into 12 mechanisms of action, not counting one DMSO control. There are actin disruptors, Aurora kinase inhibitors, DNA damage mechanisms, etc.

For this study, we would like to choose compounds that are as comparable as possible.

Firstly, compounds from different MoA groups should be selected. This way, we will probably obtain the most morphologically distinct cell phenotypes. This is not guaranteed, but it is likely. The aim of the project is not to conduct an exhaustive analysis, but to test whether, in principle, it is possible to distinguish some phenotypes of cells treated with different compounds using machine learning.

Secondly, the compounds have different concentrations, and even different concentration ranges, which also leads us to the assumption that we need to choose compounds with a similar concentration profile.

For example, the selected compounds should have at least three concentrations each, so that we can first choose a middle concentration for comparing the compounds, and then possibly expand the project to analyse all three concentrations in order to understand whether cell morphology differs depending on concentration.

We know that the dataset includes 8 concentrations for each compound. However, the authors did not leave all of them in the metadata table. Most likely, this is because the dataset authors selected the most meaningful concentrations. Therefore, it is more logical to choose from these.

In order not to make the dataset too large while testing the hypothesis, we will choose 4 compounds from different MoA groups among those that have 3 concentrations of a close order of magnitude, plus DMSO as the control.

Notable, we will not compare numerically equal concentrations of different compounds, because this would be incorrect. A concentration of 1.0 for Taxol and 1.0 for Etoposide are not necessarily biologically comparable in terms of effect strength. One compound may already have an effect at the minimum concentration, while another may require a 10 times higher concentration to produce some effect.

Therefore, the compromise chosen for the project (Table 3) is that each selected compound will have a low, medium, and high dose within that compound itself.

Table 3: Selected compounds, MoA and concentrations

MoA group	Compound	Concentrations
control	DMSO	[0.0]
cholesterol-lowering	Lovastatin	[1.5, 5.0, 15.0]
epithelial	AZ-U	[1.0, 3.0, 10.0]
DNA damage	Etoposide	[1.0, 3.0, 10.0]
Aurora kinase inhibitor	AZ258	[0.1, 0.3, 1.0]

3.3 Expectations for morphology

`DMSO` is a solvent, used as a control to avoid confusing the effect of the solvent with the effect of the drug; it is not expected to do anything and provides normal cells for comparison.

`Lovastatin` can act through the mevalonate pathway, blocking cholesterol synthesis as well as the production of small molecules needed by some proteins to function properly and attach to cell membranes. Therefore, we can expect the cells to contract, become more rounded and sparse, and lose tight contacts with neighboring cells.

`AZ-U`, as an epithelial phenotype agent, makes cells more rounded and compact and promotes closer cell-cell contacts. We expect the cells to be packed more densely into clusters, with clearer boundaries but smaller individual cell size.

`Etoposide` is a DNA-damaging compound that can ar-

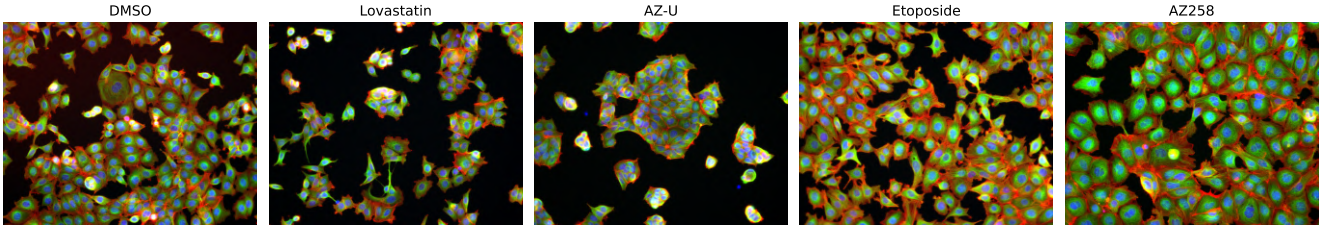


Figure 3: Example field for each compound

rest cells in the phase before division. Expected effects include changes in nuclear morphology, cell-cycle arrest or cell death in part of the population, and a possible decrease in cell density.

`AZ258` is an Aurora kinase inhibitor, disrupting the process of cell division regulated by Aurora kinase; expected effects include disruption of the mitotic spindle, chromosome segregation errors, cell-cycle arrest, apoptosis, cells becoming stuck in mitosis, rounded morphology, and abnormal nuclei.

As shown in the Figure 3, DMSO looks like a control condition with mixed morphology: many cells are well spread, cell-cell contacts are present, and there is natural variation in cell shape. Lovastatin appears more sparse; the cells more often form separate groups, with less of a dense continuous layer. The description also fits AZ-U well: dense clusters are visible, and the cells look smaller and tightly packed. For Etoposide, the description does not fully match this example field, which is still quite dense. However, 24 hours may not be enough to induce extensive apoptosis, and the compound may have mainly arrested cell division. Some cells show elongated protrusions and heterogeneous morphology, but these features are not specific enough to conclude failed cell division. They can be interpreted as changes in cell shape and cytoskeletal organization under DNA-damage stress. AZ258 generally matches the expected description: some cells appear to be in the process of division, and double or triple nuclei are visible.

Table 4: Plate coverage for compounds (bold for chosen)

Image Plate	compound_norm
Week10_40111	[AZ-U, DMSO]
Week10_40115	**[AZ-U, DMSO]**
Week10_40119	[AZ-U, DMSO]
Week3_25461	[DMSO, Etoposide]
Week3_25421	[DMSO, Etoposide]
Week3_25441	**[DMSO, Etoposide]**
Week4_27801	[AZ258, DMSO]
Week9_39206	[DMSO, Lovastatin]
Week4_27861	[AZ258, DMSO]
Week9_39221	**[DMSO, Lovastatin]**
Week9_39222	[DMSO, Lovastatin]
Week4_27821	**[AZ258, DMSO]**

3.4 Plate selection

Each plate is a separate ZIP file of about 700 MB. Downloading all 55 plates would be about 40 GB, which is not necessary for this analysis. We want to download the minimum number of ZIP archives while covering as many of the required compounds as possible (Table 4).

Therefore, we need to filter and download plates that contain the largest possible number of compounds selected for analysis at the same time.

In `BBBC021_v1_moa.csv` we cannot find the complete experiment index, but a ground truth table for mechanisms of action. It contains only those compound-concentration pairs that the dataset authors considered meaningful for MOA classification.

Some doses were not included in this MOA subset because they were inactive, overly toxic, or did not pass quality control. Since this project is built around MOA and phenotypic comparison, it is better to take the concentrations from meta data. If we used all concentrations from `BBBC021_v1_image.csv`, we could include doses that may have been unchecked for MOA analysis.

Therefore, we filter the data using the three selected concentrations.

We will also take the middle concentrations of the compounds for the main analysis later. We will compare phenotypes at one middle concentration so that the comparison between compounds is valid. This assumption is based on the idea that the dataset authors estimated the effective concentrations for the compounds in a roughly comparable way.

Nevertheless, we cannot confidently claim that all observed differences are due to the drug effect rather than plate or well effects. Hundreds of cells from the same plate or field of view cannot be treated as fully independent observations.

We will also take DMSO from all plates to check whether the controls from different plates are comparable (no batch or plate effect).

After downloading and extracting the files, we will have 52 fields of view in total: 48 treatment fields of view (4 compounds $\times$ 3 concentrations $\times$ 4 fields per compound-concentration pair) plus 4 DMSO control fields of view. This gives 52 fields of view $\times$ 3 channels = 156 TIFF files.

3.5 Image preprocessing

Each field of view consists of three TIFF files of `uint16` sized 1024x1280. After reading each channel, we:

- convert it to `float32`, to work with continuous rather than integer values and to rescale pixel intensities into a normalized range;

- normalize it to the $[0, 1]$ range by clipping the darkest 1% and brightest 1% of pixels using percentiles. Fluorescence images almost always contain outliers, such as individual extremely bright pixels, artifacts, or especially bright spots within cells. If we do not clip them, standard min-max normalization can make the rest of the image look gray. Therefore, percentile clipping is commonly used;

- build an RGB composite for visualization: red for actin, green for tubulin, and blue for DAPI. We do this after normalizing each channel separately, because if the channels are first combined into an RGB image and then normalized together, one channel may dominate the others due to differences in brightness.

3.6 Watershed baseline

Watershed is a classical segmentation method from mathematical morphology, introduced before deep learning (Beucher and Lantuéjoul, 1979). The general idea is to represent an image, or a map derived from an image, as a topographic surface. Low-value regions are treated as basins, and high-value regions are treated as barriers. If several basins are filled from different starting points, the places where these filled regions meet define watershed boundaries. In image segmentation, this produces a labelled mask: an array with the same height and width as the original image, where each pixel is assigned to one object ID, and 0 corresponds to background.

The marker-controlled watershed algorithm receives three objects: an elevation map, a set of labelled markers, and a binary mask that restricts where segmentation is allowed. In the `skimage.segmentation.watershed` implementation used here (scikit-image developers, 2025), pixels are assigned to marker basins using a priority-queue formulation related to the standard immersion-based watershed algorithm (Vincent and Soille, 1991). Pixels with lower elevation values are processed first; when several pixels have the same elevation, the order in which they enter the queue helps resolve plateaus between nearby markers. In this project, the elevation map is the negative Euclidean distance map of the cell mask. For each pixel inside the mask, the Euclidean distance transform gives the distance to the nearest background pixel. Therefore, pixels near the edge of the mask have small distance values, while pixels deeper inside the cell region have larger values. After multiplying this distance map by -1, internal regions become low-elevation basins, and watershed assigns pixels inside the mask to the labelled nuclear markers.

Pipeline:

1. The input consists of two fluorescence channels: DAPI and actin. DAPI is used to find nuclei, while actin is used to estimate the broader whole-cell area.

2. Both channels are smoothed with a Gaussian filter. This reduces pixel-level noise and makes the following thresholding step more stable.

3. Otsu thresholding is applied to the smoothed DAPI image (Otsu, 1979). Otsu is an automatic threshold-selection method: it computes one intensity threshold for the image, and pixels above this threshold are treated as foreground. This produces a binary nuclei mask.

4. The nuclei mask is cleaned morphologically. Small bright artefacts are removed, and small gaps inside nuclei are closed. This makes the nuclear objects more stable before marker detection.

5. For each pixel inside the nuclei mask, the Euclidean distance to the nearest background pixel is calculated. This produces a nuclear distance map. Pixels near nuclear borders have low values; pixels closer to nuclear centers have high values.

6. Local maxima are detected on the nuclear distance map. These maxima correspond to approximate nuclear centers. Their coordinates are converted into a binary peak mask and then labelled. The resulting labelled objects are the seed markers for watershed: ideally, one marker corresponds to one nucleus and therefore to one cell.

7. Otsu thresholding is also applied to the smoothed actin image. This produces an actin-based mask of the broader cell-containing region.

8. The actin-based mask is combined with the nuclei mask using a logical OR operation. A pixel is included in the final cell mask if it belongs either to the actin mask or to the nuclei mask. This prevents detected nuclei from being excluded when the actin signal is weak or fragmented around them.

9. For each pixel inside this combined cell mask, the Euclidean distance to the nearest background pixel is calculated again. This produces a distance map of the full cell-containing region.

10. This full-cell distance map is multiplied by -1 and used as the watershed elevation map. The labelled DAPI-derived nuclear markers define the starting basins, and the combined actin+nuclei mask restricts the segmentation to likely cellular pixels.

11. Watershed assigns every pixel inside the combined mask to one of the nuclear markers according to the elevation map. If several nuclei lie inside one connected actin-positive region, watershed attempts to divide that shared region between the markers. The output is a labelled whole-cell mask, where each non-zero label corresponds to one predicted cell.

The main limitation of this approach is that the final boundaries are determined by thresholded masks, marker placement, and distance-map geometry, not by an actual biological understanding of cell shape. As we

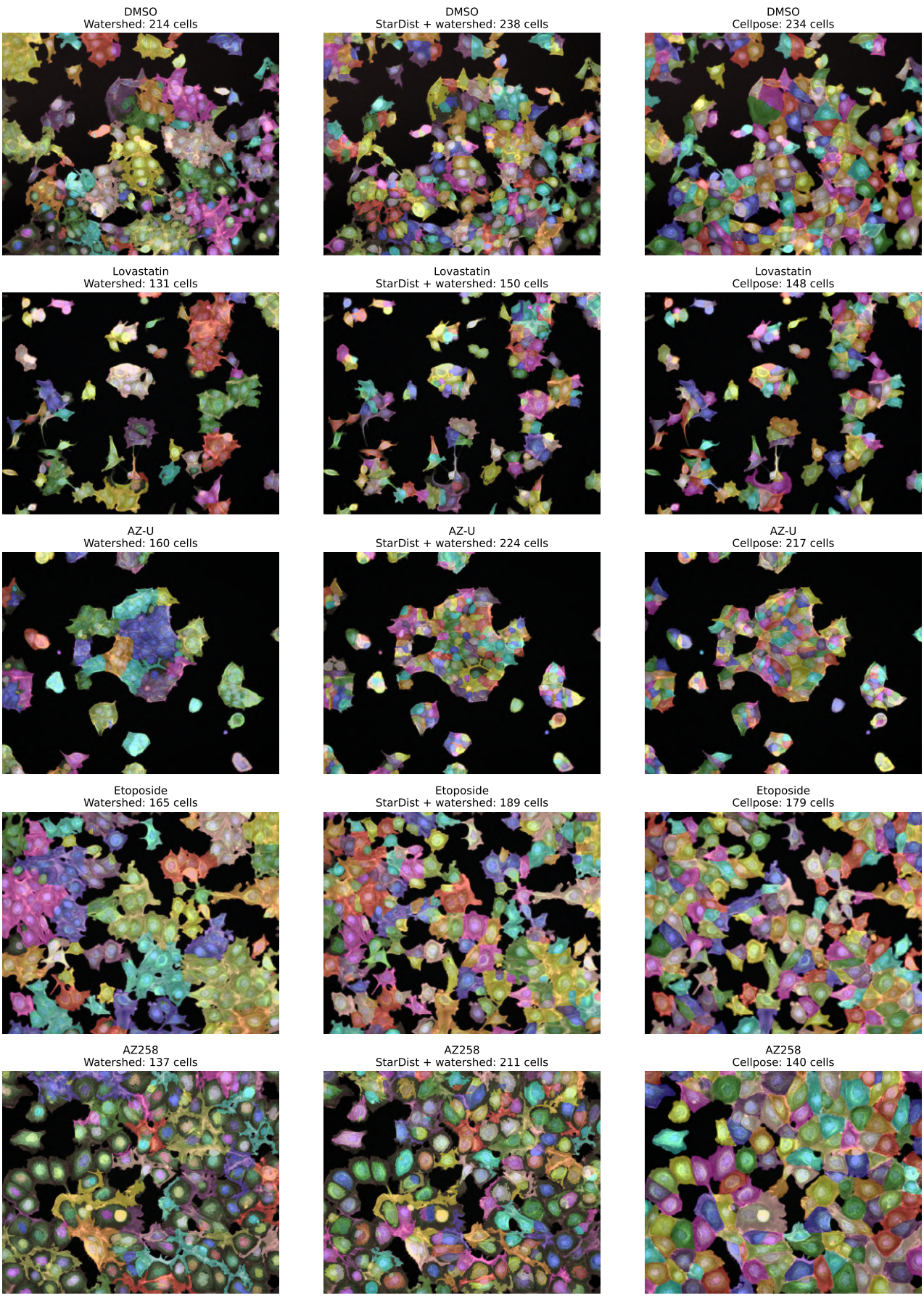


Figure 4: Comparison of Watershed, StarDist and Cellpose

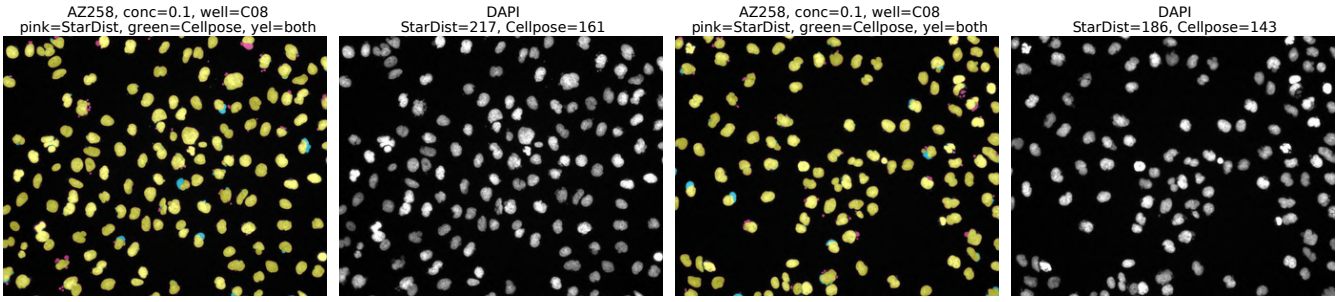


Figure 5: Nuclei segmentation comparison between Stardist and Cellpose for AZ258

see on Figure 4, watershed can work well for isolated cells, but in dense clusters it often fails: missed nuclear markers can merge cells, extra markers can split one cell into several objects, and poor actin masks can force boundaries into biologically incorrect positions.

3.7 StarDist

StarDist is a deep-learning method for instance segmentation of approximately star-convex objects, especially cell nuclei in microscopy images (Schmidt et al., 2018). Unlike semantic segmentation methods that first classify pixels and then group them into instances, StarDist directly predicts object candidates. Each object is represented as a star-convex polygon: for a point inside the object, the boundary is described by radial distances along a fixed set of directions. This representation is well suited for nuclei, which are usually compact and approximately round or elliptical, but it is less suitable for whole-cell segmentation in our images because cytoplasm and cytoskeletal structures can be irregular, elongated, fragmented, and overlapping.

For each pixel, StarDist predicts two outputs. The first output is an object probability, indicating how likely this pixel is to belong to an object and to serve as a good object center. In the original method, the training target for this probability is not just a binary foreground label, but a normalized Euclidean distance to the nearest background pixel, so pixels closer to object centers receive higher target values (Schmidt et al., 2018). The second output is a vector of radial distances where each value is the predicted distance from pixel to the object boundary along one of predefined directions. These distances define a polygon candidate centered at the pixel. During prediction, many pixels can propose similar polygons for the same nucleus. Therefore, StarDist keeps only candidates whose object probability is above a probability threshold and then applies non-maximum suppression (NMS): overlapping polygon candidates are compared, and lower-scoring redundant candidates are removed (Schmidt et al., 2018). The output is an instance-labelled nuclei mask, where 0 is background and each positive integer label corresponds to one detected nucleus.

In this project, the pretrained `2D_versatile_fluo` StarDist model is applied to the normalized DAPI channel. The DAPI channel is used because it directly stains DNA and therefore provides the clearest signal for nuclei. We use StarDist in two ways. First, the number of StarDist-detected nuclei is compared with the number of nuclei detected by Cellpose to check whether the cell-count estimates have the same order of magnitude. Second, the StarDist nuclei labels are used as markers for a marker-controlled watershed. This tests whether replacing heuristic DAPI markers with deep-learning-based nuclear instance markers improves whole-cell segmentation on Figure 4.

Pipeline:

1. The input consists of three channels: DAPI, actin, and tubulin. DAPI is used for nuclei detection, while actin and tubulin are used to estimate the broader whole-cell area.

2. The DAPI channel is normalized and passed to the pretrained StarDist model. StarDist returns an instance-labelled nuclei mask: background pixels have value 0, and each detected nucleus receives its own integer label $1, 2, \ldots, N$.

3. For nuclei counting, the StarDist labels are used directly.

4. For StarDist-guided watershed, the StarDist nuclei labels are converted to watershed markers. Unlike naive watershed, no search of markers is needed here because StarDist has already produced instance-level nuclear objects.

5. Actin and tubulin channels are smoothed with a Gaussian filter, normalized, and combined using a pixel-wise maximum (logical OR). signal, where a pixel can contribute to the cell mask if either actin or tubulin shows signal.

6. Otsu thresholding is applied to the combined actin+tubulin signal (Otsu, 1979). This produces an approximate whole-cell mask.

7. The whole-cell mask is combined with the StarDist nuclei mask using a logical OR operation. This ensures that detected nuclei remain inside the allowed segmentation region even if the cytoskeletal signal is weak or fragmented around them.

8. As in the watershed baseline, small disconnected objects are removed from the combined mask, a Euclidean distance map is calculated inside this mask, and its negative version is used as the watershed

elevation map. The difference is that the markers come from StarDist nuclei labels, and the allowed cell mask is built from actin+tubulin+nuclei rather than actin+nuclei.

9. Watershed assigns every pixel inside the combined mask to one of the StarDist nuclear markers according to the elevation map. The output is a labelled whole-cell mask, where each non-zero label corresponds to one predicted cell associated with one StarDist-detected nucleus.

This hybrid method is still not a fully learned whole-cell segmentation model. StarDist only provides nuclear instance markers; the final cell boundaries are still determined by the thresholded actin+tubulin mask and the geometry of the distance transform used by watershed. Therefore, the method can improve marker placement compared with naive watershed, but it can still fail when nuclei are missed, when one nucleus is split into several StarDist instances, or when the cytoskeletal mask does not reflect true cell boundaries in dense clusters on Figure 4.

Since we do not have ground truth for cells, we try to indirectly check whether Cellpose and StarDist at least detect cell nuclei comparably (Figure 5). This is not a full validation, because it does not prove that the masks are equally good. After all, for whole-cell segmentation we provide different input channels. Still, a similar number of objects would mean that Cellpose nuclei does not fail completely in terms of count relative to StarDist.

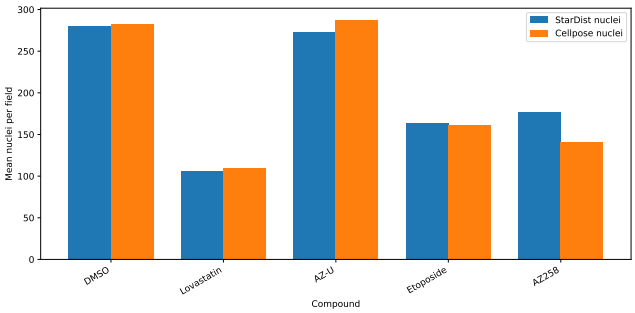


Figure 6: Nuclei count for Stardist and Cellpose

As shown in Figure 6 and Table 5, the number of segmented nuclei is almost the same for both models in most cases. A small difference is observed for AZ-U, and a stronger difference is observed for AZ258, with an average difference of 37 cells.

To understand the source of this discrepancy, we inspect the AZ258 segmentation examples in Figure 5. StarDist tends to oversegment very small nuclear fragments, which may not correspond to complete cells. At the same time, compared with Cellpose, StarDist misses some larger nuclei attached to neighbouring nuclei. This is a more serious undersegmentation error, because these missed nuclei are often surrounded by cytoplasm, and the corresponding structures can reasonably be counted as cells.

Table 5: Nuclei count for Stardist and Cellpose

Compound	Stardist nuclei	Cellpose nuclei
DMSO	280.2	282.2
Lovastatin	106.4	110.1
AZ-U	272.9	287.2
Etoposide	163.4	161.7
AZ258	177.3	140.4

3.8 Cellpose

Cellpose is a deep-learning-based segmentation algorithm pre-trained on highly diverse microscopy. Its diverse manually annotated dataset contained more than 70,000 segmented objects (Stringer et al., 2021).

The model is based on the U-Net architecture, a widely used neural network for biomedical image analysis. It first compresses the image through downsampling to extract global context and then restores it through upsampling back to the original resolution to define precise object boundaries. The authors added residual blocks and image style representation: skip connections are used between layers of the same size, together with global skip connections from image styles computed at the lowest resolution to all subsequent computational steps.

To transform manually annotated masks into a vector flow representation that can be predicted by the neural network, the authors use a simulated diffusion process starting from the centre of the mask. This produces spatial gradients pointing toward the centre of the cell, including indirect paths around corners and complex shapes. The x and y gradients are combined into a single direction from 0° to 360°.

The neural network is trained to predict horizontal and vertical gradients, as well as whether each pixel belongs to a cell. The three predicted maps are combined into a single gradient vector field.

At test time, the predicted gradient vector fields are used to construct a dynamical system with fixed points whose basins of attraction represent the predicted masks. In simple terms, each pixel follows the gradients toward its final fixed point.

All pixels that converge to the same fixed point are assigned to the same mask. This solves the main problem of watershed: touching cells are separated because their centers are different, and the vectors from them point in different directions.

Cellpose is a family of convolutional neural networks. The `cyto3` model is a general-purpose version that works across a wide range of cell types without further training. It expects a two-channel image as input. For BBBC021, this is:

- **cytoplasm** – actin, which shows the overall cell shape;
- **nuclei** – DAPI.

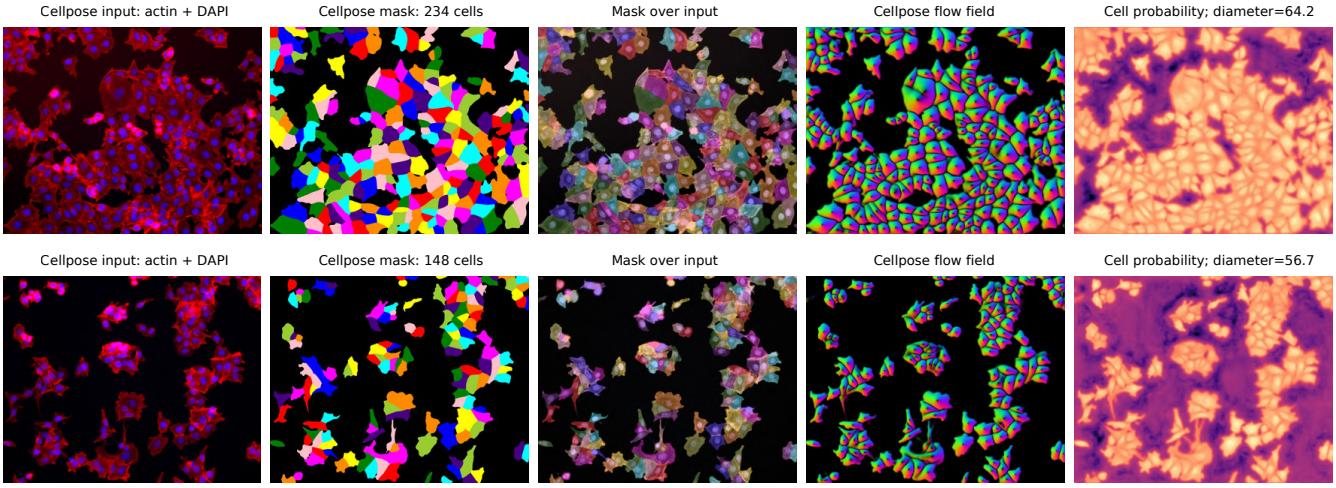


Figure 7: Cellpose flows and cell probability (DMSO in the first line, Lovastatin in the second line)

3.9 Cellpose hyperparameter tuning

According to the Cellpose v3.1 documentation, the default parameters are:

- `diameter=None` (auto-estimation)
- `flow_threshold=0.4`
- `cellprob_threshold=0.0`

Hyperparameters (Cellpose developers)

`diameter`, in pixels

Cellpose uses the diameter to rescale the image before segmentation. The v3 cytoplasm models were trained with objects rescaled to a mean diameter of 30 pixels, while the nuclei model was trained with a mean diameter of 17 pixels. Therefore, `diameter` controls the scale at which the network sees the objects.

- `diameter = None` or 0: Cellpose estimates the object size automatically.
- `diameter > 0`: the expected object diameter is specified manually.

Automatic diameter estimation is a two-step procedure. Cellpose first runs the image through the network and obtains a style vector, then uses a linear regression model to estimate object size. The image is resized according to this estimate, Cellpose is run again, and the final diameter is refined from the median diameter of the predicted ROIs. If the specified diameter is too small, Cellpose may over-split cells; if it is too large, neighbouring cells may be merged.

`cellprob_threshold`

Default: `0.0`.

Cellpose predicts three maps: horizontal flows, vertical flows, and cell probability. The cell probability output is a logit. These logits are inputs to a sigmoid centered at zero and typically range approximately from `-6` to `+6`. Pixels with values above `cellprob_threshold` are used in the flow dynamics step to construct ROIs.

- Lower values make the model more permissive and can recover dim or weakly stained cells.
- Higher values make the model stricter and can reduce masks from dim regions or background texture.
- Too low a value may include debris or weak artefacts; too high a value may remove real but low-contrast cells.

`flow_threshold`

Default: `0.4`.

This parameter is a mask-quality filter. After Cellpose predicts flows and constructs candidate masks, it recomputes flow gradients from the recovered masks and compares them with the flows predicted by the neural network. The comparison is based on mean squared error. The `flow_threshold` parameter is the maximum allowed flow error for a mask to be kept.

- Lower values are stricter: more masks with inconsistent or distorted flows are discarded.
- Higher values are more permissive: more ROIs are retained, but some may have poor or biologically implausible shapes.
- Increasing `flow_threshold` can help if Cellpose returns fewer ROIs than expected; decreasing it can help if too many ill-shaped ROIs are accepted.

3.10 Grid search with set diameter

To optimize the Cellpose configuration, a grid search (Figure 8, Figure 9) was performed around the default settings. The baseline configuration used automatic diameter estimation, `cellprob_threshold=0.0`, and `flow_threshold=0.4`. For the grid search, `diameter` was tested as `None`, 50, 70, and 90 pixels. These values were chosen because a preliminary Cellpose run with

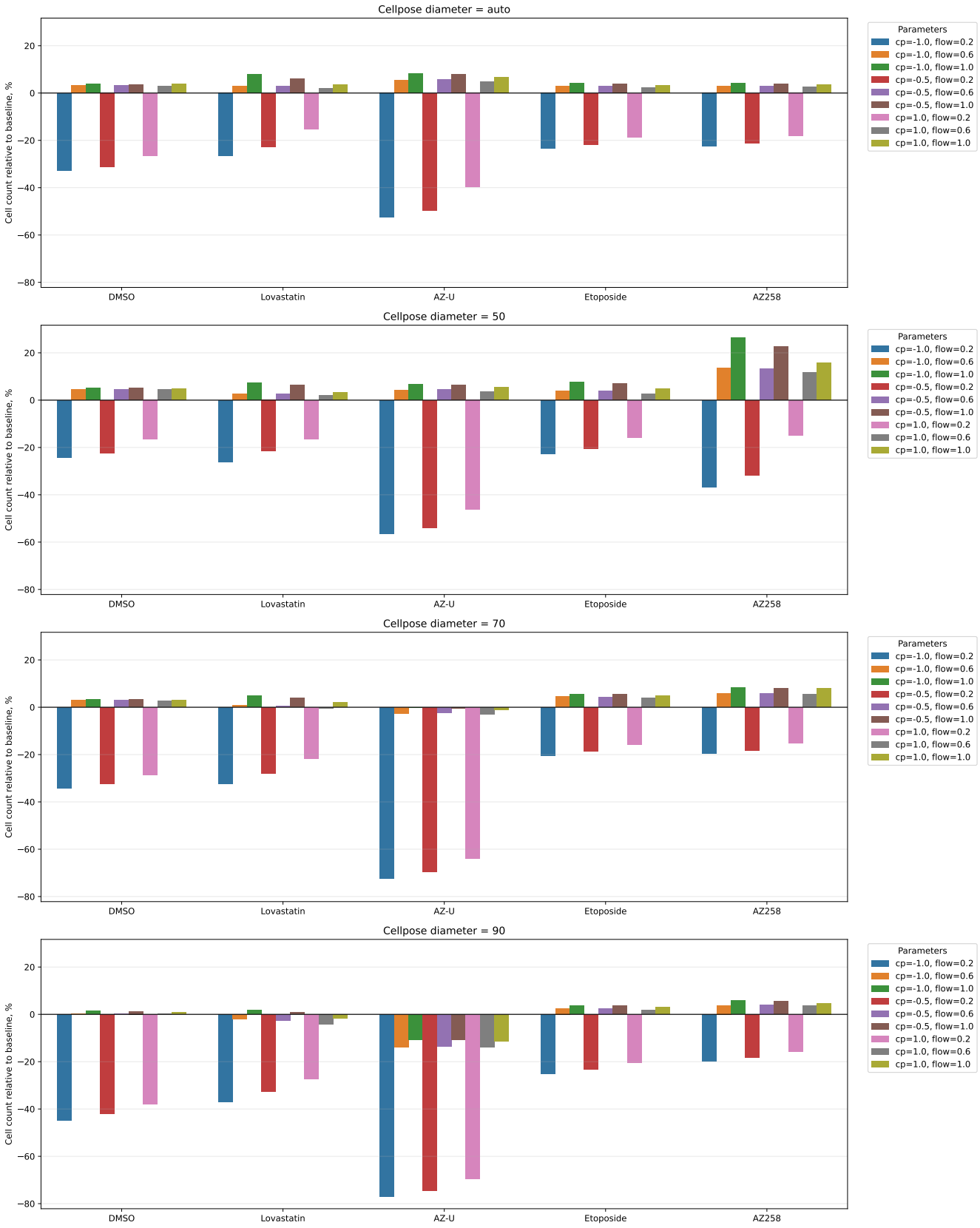


Figure 8: Cell count during Cellpose grid search

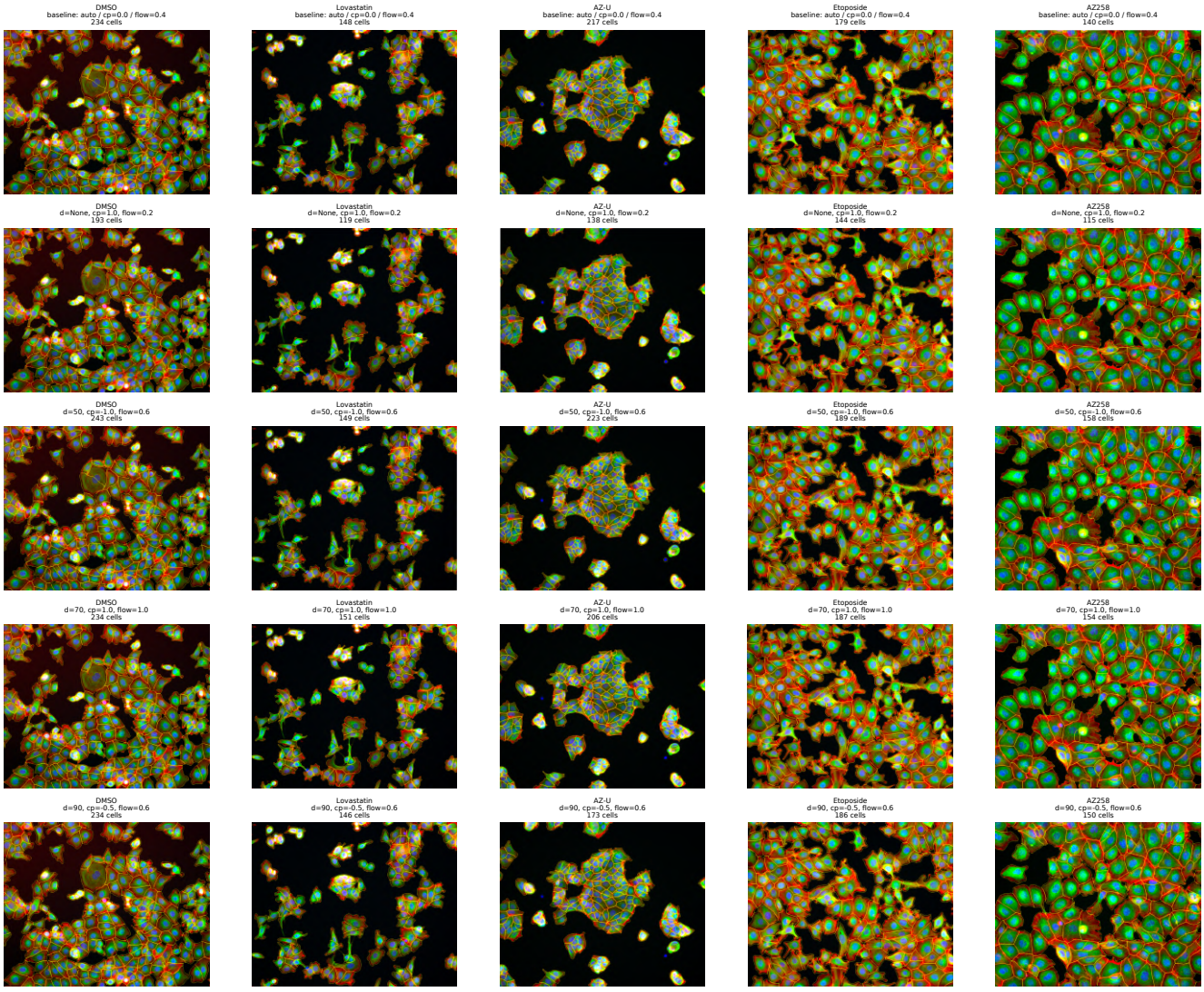


Figure 9: Cellpose grid search overlays with manual diameter estimation (full figure on Drive)

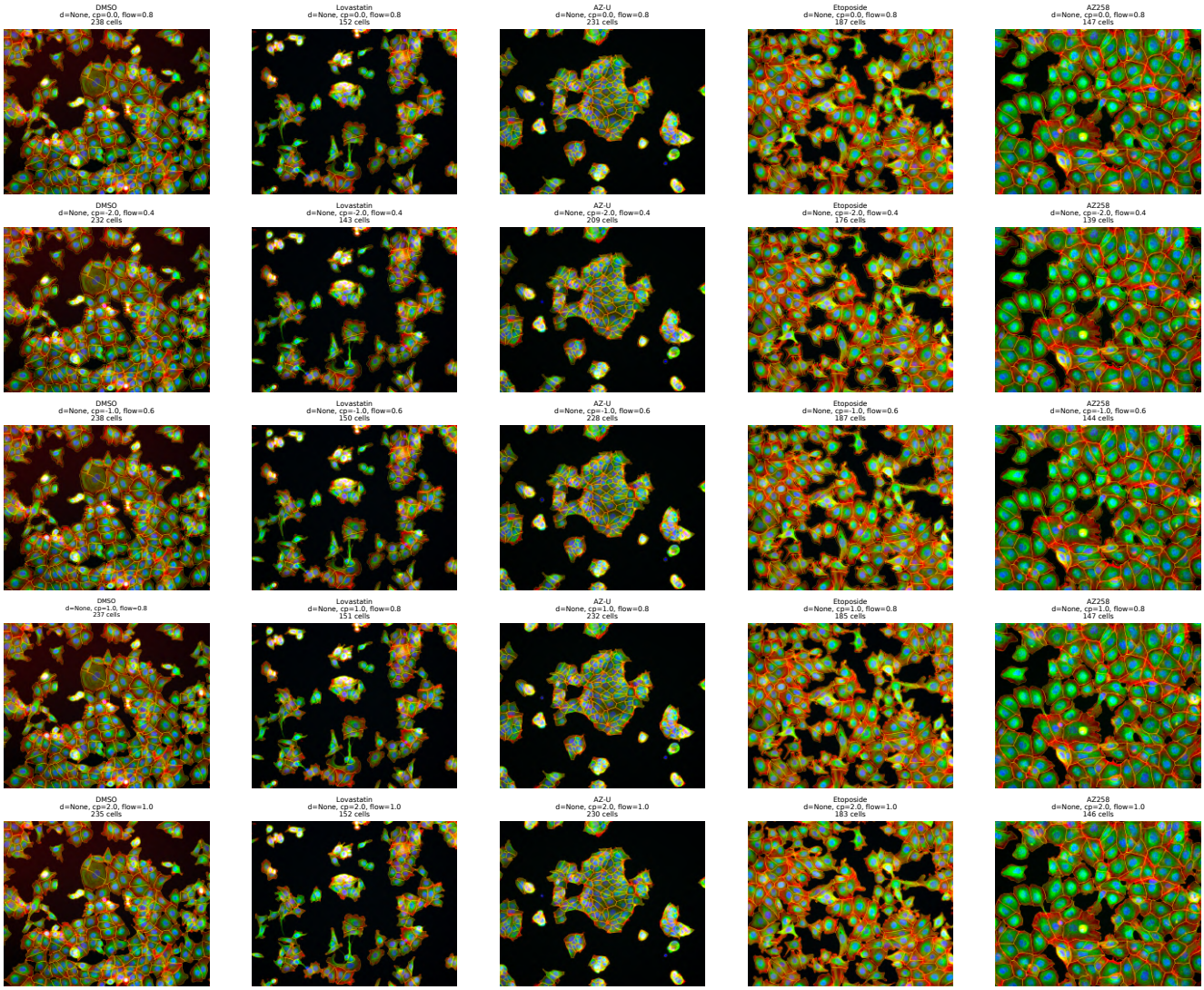


Figure 10: Cellpose grid search overlays with automatic diameter estimation (full figure on Drive)

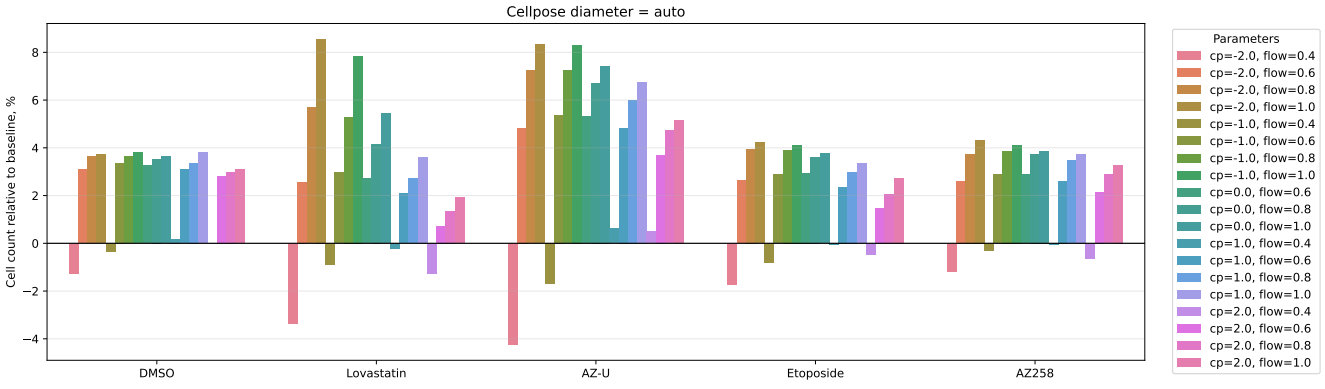


Figure 11: Cell count during Cellpose grid search with automatic diameter

automatic diameter estimation returned average cell diameters between approximately 43 and 96 pixels for different compounds, so the tested manual diameters covered this observed scale range. The `flow_threshold` values 0.2, 0.6, and 1.0 were selected to test settings below and above the default value of 0.4. The `cellprob_threshold` values -1.0, -0.5, and 1.0 were selected to cover a range around the default value of 0.0, from more permissive to stricter foreground acceptance. Each parameter combination was applied to the same selected fields, and the resulting masks were compared by object counts and visual overlay inspection.

On Figure 8, the diagram shows the change in the number of detected objects relative to the model with default parameters. A high bar does not always mean better performance: it can reflect either true recovery of missed cells or over-segmentation.

The main conclusion is that `flow=0.2` should be excluded. In all panels and for all compounds, the bars for `flow=0.2` go strongly downward. At the same time, visual inspection of the masks shows that whole cell regions and clusters disappear. This looks like overly strict mask filtering by flow quality: complex, dim, merged, or poorly separable cells receive unstable flows and are excluded.

Effect of diameter

Different compounds respond to `diameter` differently because the real cell sizes are very different.

AZ-U cells are the smallest ones (`Cellpose mean diameter = 43.0`). They are gathered into compact clusters, have blurred boundaries and merged nuclei, sometimes with 3–5 nuclei in one clump, making it difficult to determine where one cell ends and another begins. They are difficult to segment. Therefore, with `d=50`, segmentation is generally acceptable, but some small isolated cells are lost. With `d=70`, and especially with `d=90`, the model starts merging several cells into one and loses cells.

Lovastatin cells are slightly larger (`Cellpose mean diameter = 56.7`), but very heterogeneous: some are merged with unclear nuclei and boundaries, while others are large and have long protrusions. They form clusters, but fewer than AZ-U and the other compounds; they are more sparse, but not isolated. This makes them difficult to segment. Therefore, with `d=50`, they are generally segmented reasonably well, but small cells are lost because the fixed diameter is too large for them. With `d=70` and above, clear under-segmentation appears.

DMSO cells are medium-sized (`Cellpose mean diameter = 64.2`). They are heterogeneous, with both smaller and larger cells, and they are quite densely packed. However, nuclei and boundaries are not always clear, and there are no large protrusions. This makes segmentation moderately difficult. For this reason, the effect of diameter is relatively weak for DMSO.

Etoposide cells are larger mainly because of protrusions (`Cellpose mean diameter = 80.7`). They lie close to each other, but their boundaries are blurred. They have protrusions that the model identifies as separate cells, and none of the tested Cellpose parameters can fully solve this. However, the nuclei are clear, so the overall segmentation difficulty is moderate. They are segmented well with `d=70/90`, while with `d=50` the cells are slightly fragmented.

AZ258 cells are the largest in the dataset (`Cellpose mean diameter = 96.3`). They mostly lie close to each other, have fairly clear contours and large nuclei, and are relatively convenient for segmentation. For them, `d=50/70` are clearly too small and increase the number of segmented cells by cutting large cells into pieces or by narrowing mask boundaries so that they do not reach the real cell borders. At the same time, even `d=90` is not ideal, because a large dim cell is clearly undersegmented or split in half. This shows that with a fixed diameter, information about very large but real cells is lost.

Therefore, a fixed diameter for all compounds is not suitable here. `d=50` does not strongly harm small or medium cells, but does not allow normal segmentation of AZ258 and Etoposide. `d=70/90` do not strongly harm large AZ258 and Etoposide cells, but cause undersegmentation of Lovastatin and AZ-U. The most stable option for the whole dataset is `diameter=None`, i.e. auto-estimation.

Other parameters

`cellprob_threshold` is a fine sensitivity adjustment. In practice, as we see from the masks, it affects the margin around the segmented object. With `cp=-0.5` and `cp=-1.0`, more background is captured, while with `cp=0` and `cp=1`, the mask boundary fits the cell more tightly. This is because this parameter is the probability/confidence threshold used by Cellpose to decide which pixels are considered cellular and included in the mask.

Based on visual inspection of the masks, with `cp=-0.5` or `cp=-1.0`, the threshold is lower, so Cellpose includes more weak or boundary pixels. Therefore, the masks often become larger and dim cells may appear, but the risk of capturing background, cytoplasmic fragments, and imaging artefacts also increases. For our task, negative values are better avoided: for Lovastatin, they lead to over-segmentation of artefacts and cell parts that clearly should not have been included in the masks, with pieces of cytoplasm segmented as separate small cells. For Etoposide, they make the mask boundaries too large for the cells. Whether to use `cp=0` or positive values will be clarified in the next grid search.

The difference between `flow_threshold=0.6` and `flow_threshold=1.0` is moderate, but `flow_threshold=1.0` more often keeps ovefitting masks. For morphometry, this is riskier than a small undercount.

In principle, it would be safe either to keep the baseline or to slightly adjust the flow to `flow_threshold=0.6`, because for two compounds, the default settings lose 1–2 cell parts: Etoposide and AZ-U. For the other compounds, the baseline is fine.

For fine-tuning, we will perform a second grid search.

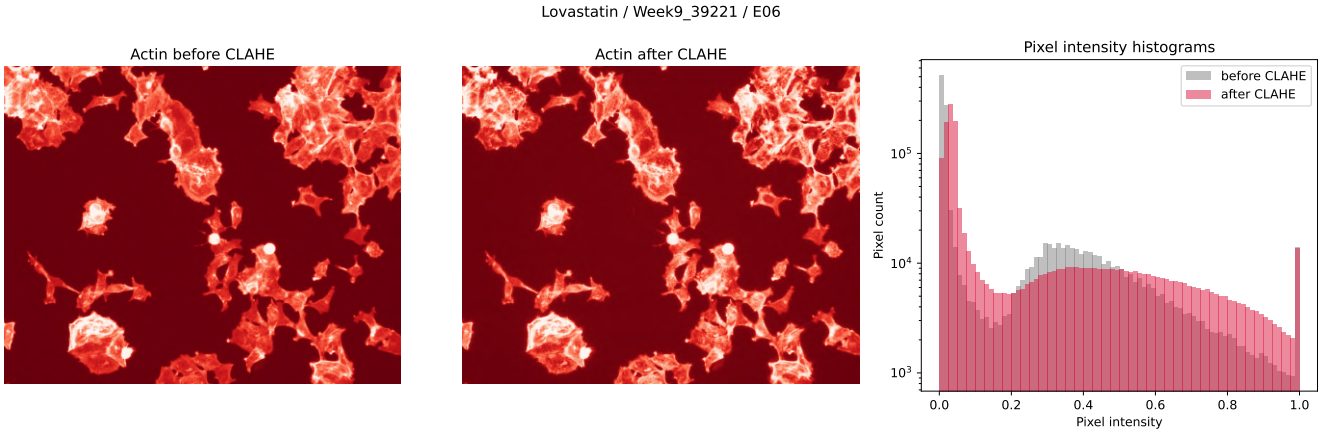


Figure 12: Actin channel before and after CLAHE

3.11 Grid search (automatic diameter)

The second grid search (Figure 11, Figure 10) was designed as a refinement of the first one. The first grid showed that manually fixing `diameter` is not a robust strategy for this dataset: the apparent cell diameter varies strongly between compounds, because different treatments affect cell size and morphology in different ways. Since the final pipeline should work across compounds with different phenotypic effects, a single manually selected diameter would either oversplit larger cells or merge smaller neighbouring cells. Therefore, in the second grid search, `diameter` was kept automatic for all variants, and the search focused on `cellprob_threshold` and `flow_threshold`. The value `flow_threshold=0.2` was also removed, because in the first grid search it performed poorly and caused entire clusters of cells to be lost. The second grid tested `flow_threshold` values 0.4, 0.6, 0.8, and 1.0, covering the default setting and more permissive mask-quality filtering. The `cellprob_threshold` range was expanded to -2.0, -1.0, 0.0, 1.0, and 2.0, because the first grid did not clearly identify which threshold was best. As before, all variants were evaluated on the same selected fields using object counts and visual overlay inspection.

On Figure 11, we see more subtle and sometimes even insignificant changes in the masks caused by parameter changes. Most bars lie within 4% relative to the baseline, while the others are up to approximately 8%. Therefore, here we are testing the sensitivity of the model to `cellprob_threshold` and `flow_threshold`.

The main conclusion is that `diameter=None` works robustly for all compounds, and further parameter tuning gives relatively small changes in cell count.

For DMSO, `cp` and `flow` almost do not change the result. Visually (Figure 10), there are no sharp changes in segmentation. Slight over-segmentation starts only at the very end, at `cp=2.0, flow=1.0`. At `cp=-2.0`, it is visible that the margins of the masks from the real cell boundaries become larger, but overall the segmentation remains acceptable.

For AZ258, the absence of a strong effect is even more pronounced: in general, all parameters are acceptable, because the cells are large, clear, high-contrast, and do not have many protrusions. This is interesting that even the mask boundaries and cell boundaries generally match well, because most cells are densely packed.

For Lovastatin, `cp=-2.0, flow=0.4` is definitely unsuitable because cells are lost. After that, segmentation is normal when changing parameters up to `cp<1.0`. With `cp>=1.0`, the masks become tighter and tighter and eventually start cutting off cell protrusions. This is probably problematic here because some cells have long protrusions. `flow` does not play a major role, except that it compensates for the effect of `cp=-2.0`, and at `flow>=0.6` cells are no longer lost.

AZ-U is a difficult case. The baseline with default parameters is definitely unsuitable for it, as are some negative `cp` values and even `cp=1.0, flow=0.4`, because of under-segmentation. At `cp>=2.0`, part of the cells is already cut off and very small cells are created from cytoplasm. Therefore, the working zone for AZ-U lies at the intersection of `flow=0.6/0.8` and `cp=-2.0/-1.0/0.0/1.0`, although there is still slight under-segmentation and slight over-segmentation of small cell fragments.

For Etoposide, the default parameters are also unsuitable because cells are lost. However, the other parameters are acceptable, although negative `cp` values expand the mask boundaries compared with the cell boundaries. The most optimal choice is at the intersection of `flow=0.6/0.8` and `cp=0.0-2`.

Increasing it to `flow=0.6` or `flow=0.8` often gives the best balance: it recovers some objects, but does not yet preserve too many extra masks. `flow=1.0` gives the maximum number of cells, but the masks show that part of this increase may be over-segmentation.

For `cellprob_threshold`: its negative values often lose cells at `flow=0.4`, while at higher `flow` they produce masks that are too wide. Conversely, `cp=2.0` makes the masks too narrow and cuts off parts of cells. Therefore, conservatively, we will keep `cp=0.0`, but increase the flow threshold to `0.8`. This should help Etoposide and AZ-U, where the baseline loses individual cells or parts of cells, but should not strongly worsen segmentation for the other compounds.

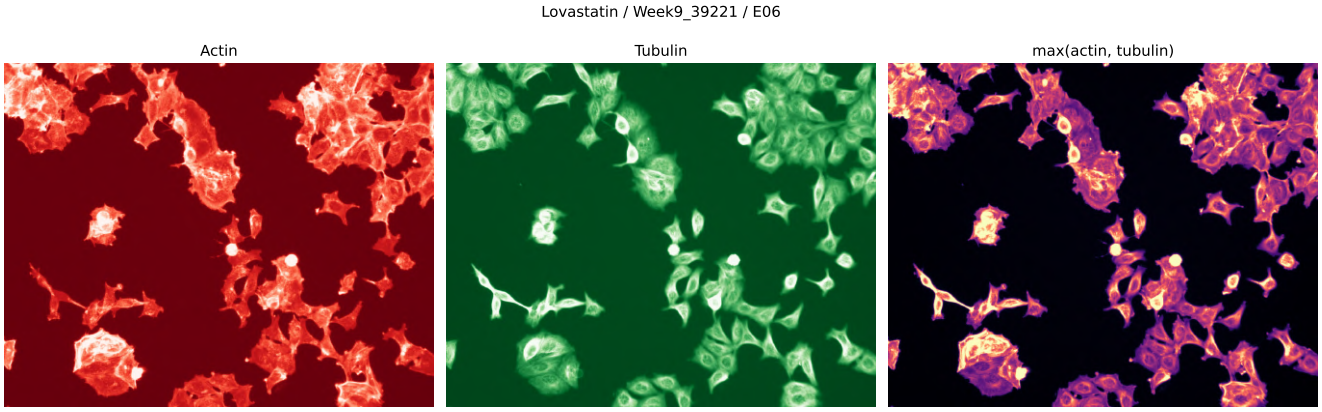


Figure 13: Merged actin and tubulin channel

3.12 Improving Cellpose segmentation

In addition to tuning hyperparameters, such as the expected cell diameter, the confidence threshold `cellprob_threshold`, and `flow_threshold`, we can also try to further improve the segmentation. The segmentation is already good, but not perfect, especially for AZ-U, where under- and oversegmentation remain at any parameter settings. Compared with Watershed, this is not a serious problem, but we will still try to improve the segmentation. We will try to make the input data more complete and higher-contrast using:

1. **Merging actin and tubulin into one cytoplasmic channel** using `np.maximum` (Figure 13). Both proteins are located in the cytoplasm; when one of them is depolymerized, the other can partially compensate for the signal. Logically, this works like an "OR" operation between the two signals. We merge these channels both without CLAHE and after CLAHE, to compare whether there is a combined effect.

2. **CLAHE (Contrast-Limited Adaptive Histogram Equalization)** – local contrast enhancement (Figure 12). This method divides the image into small regions and normalizes the contrast separately within each region, while limiting the amplification so that noise does not increase too much. This is a standard method in medical imaging, including ultrasound, X-ray images, and histology.

Both methods, together with the tuned hyperparameters, can complement each other.

For CLAHE, we will use `equalize_adapthist`. It has two parameters:

- `kernel_size=128` – the size of the local window. The smaller the window, the more aggressive the local normalization. We choose a default parameter equal to 1/8 of 1024, which fits the size of our images.

- `clip_limit=0.01` – limits the maximum contrast amplification within each local region. Values are normalized between 0 and 1; higher values give more contrast.

We applied CLAHE and channel merging separately, compared them with the tuned model, and also tested a combined approach in which all three strategies were used together with the hyperparameter-tuned model (Figures 14 and 15).

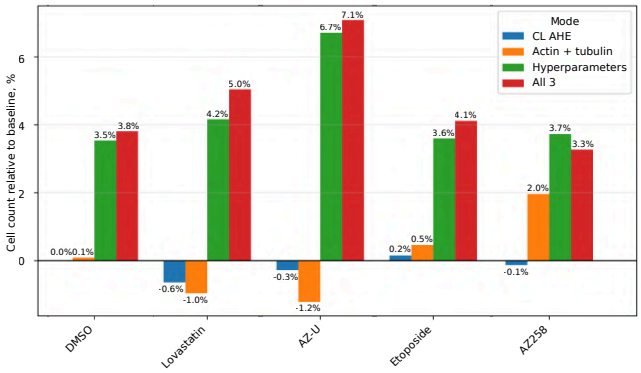


Figure 14: Cellpose enhanced modes cell count

As a result, applying all three methods together led to oversegmentation. The model started to identify structures as separate cells that were not segmented as separate objects when the methods were applied individually. These were mostly cytoplasmic fragments. Segmentation was especially poor for Etoposide: the masks split cells into several parts while also undersegmenting large, clearly visible cytoplasmic regions. However, when used separately, none of the methods produced poor segmentation for Etoposide; the problem appeared only when they were combined. AZ258 was also affected by this synergy: the masks became much smaller than the actual cell boundaries. Most likely, this happened because the simultaneous contrast enhancement from CLAHE and the merging of tubulin with actin created too many high-contrast boundaries within a single cell. Cellpose then started to interpret these internal structures as boundaries between separate cells.

It is also interesting that the channel merging completely failed for Lovastatin. At least five cells that were clearly visible by eye were not segmented in the field of view. For the other compounds, the changes were not

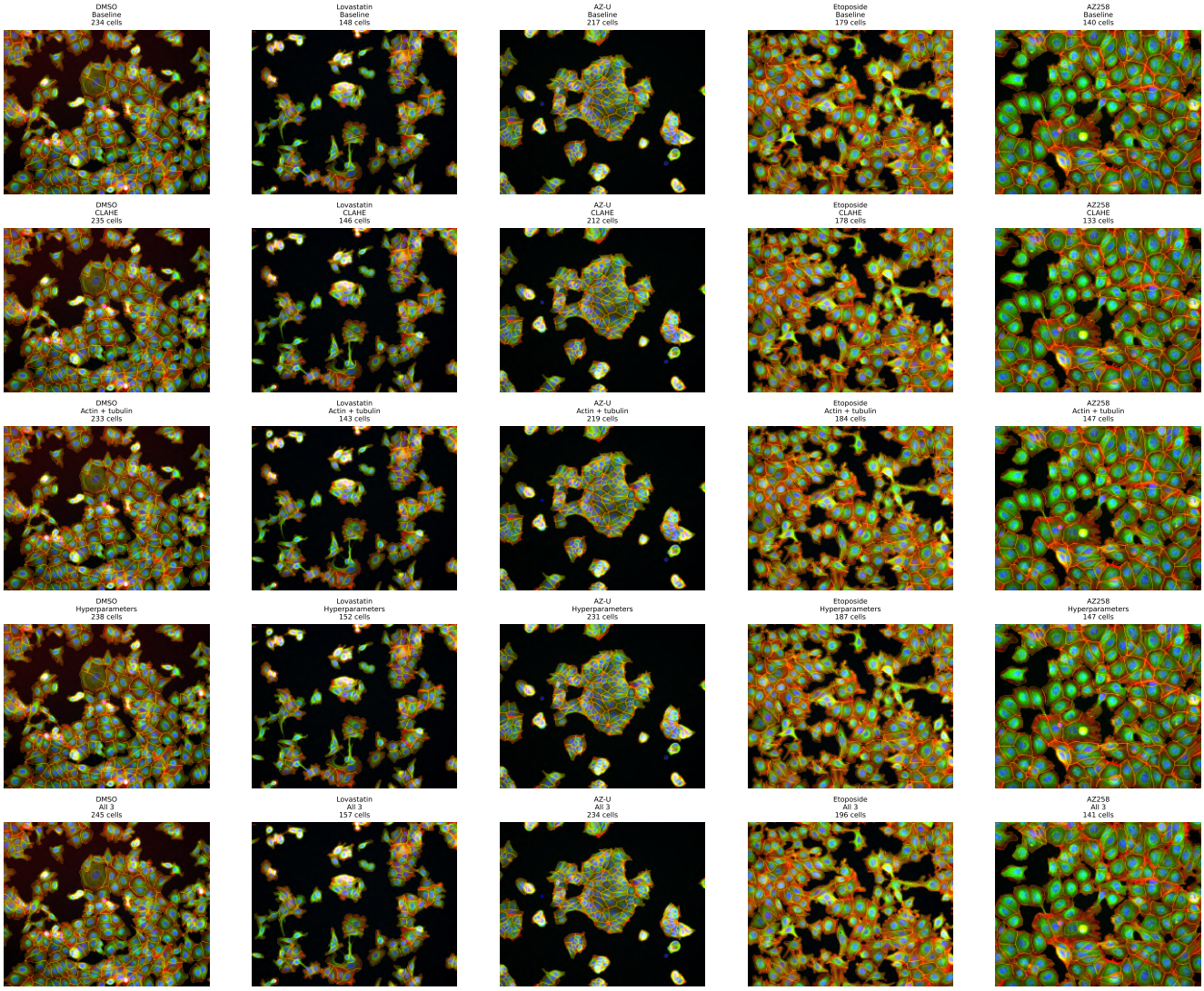


Figure 15: Comparison of Cellpose with CLAHE, merged channels and tuned parameters

substantial based on visual inspection.

CLAHE alone was also not very poor, but it was not fully reliable. It slightly improved segmentation for DMSO and AZ258, but for Lovastatin and AZ-U it missed cells.

3.13 Segmentation quality control

After three grid searches for Cellpose parameter tuning and input improvement, we obtained a model that was used to generate cell masks for all five compounds at the middle concentration. Now it is important to check whether our expectations were met:

- whether the plates are similar in the number of segmented cells;
- whether there are outliers and what they are related to;
- whether DMSO from different plates looks similar;
- whether the cell counts agree with the overlay masks.

In Table 6, we see the mean number of segmented cells per field and the corresponding standard deviation. AZ258 is the most consistent condition: the difference between fields is about 10 cells. Etoposide varies slightly more, while DMSO and Lovastatin are of a similar order. AZ-U is concerning because its standard deviation is large, approximately half of its mean cell count.

Table 6: Summary statistics of cell counts

Compound	n fields	mean cells	std cells	total cells
AZ-U	4	297.2	104.4	1189
AZ258	4	134.8	10.3	539
DMSO	4	285.2	44.4	1141
Etoposide	4	179.5	19.6	718
Lovastatin	4	103.5	41.4	414

We check this in Figure 16. AZ-U has a strong outlier, which explains its large standard deviation. However, the other three AZ-U fields are close to each other, which keeps the median more stable. We will not remove this outlier. Instead, we check the original image to see whether this field really contains so many cells or whether this is a segmentation error. As shown in Figure 17, the mask overlay confirms that there are indeed many cells in this field, approximately twice as many

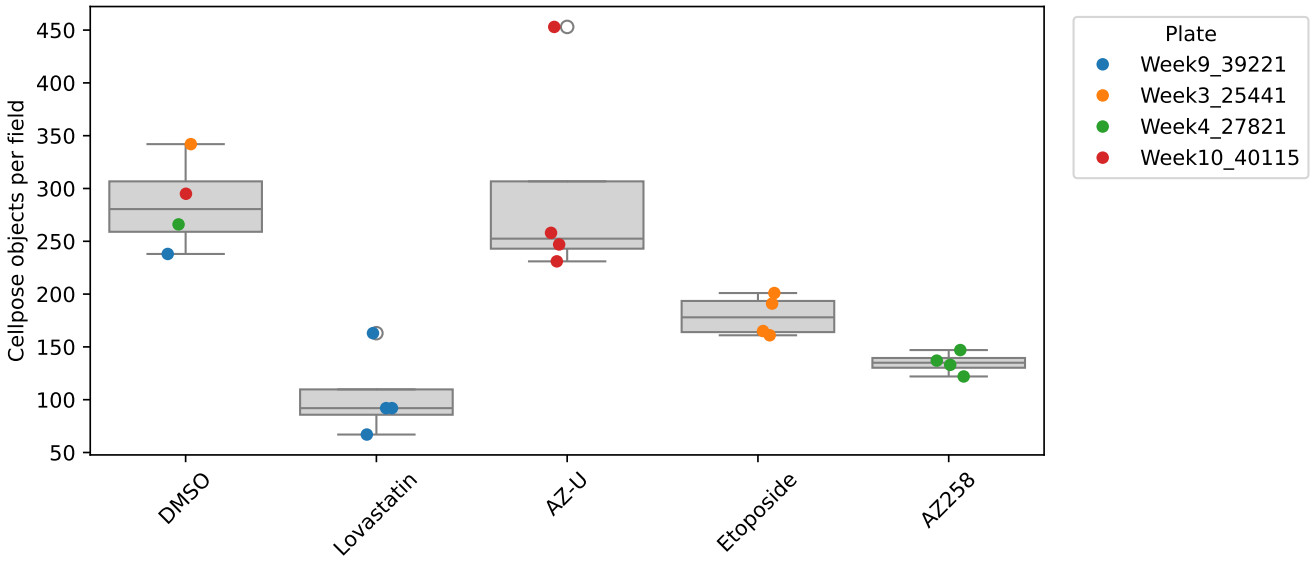


Figure 16: Cell counts per field by compound and plate

as usual. Several cell clusters are located very close to each other. This is not a segmentation error, but a specific property of this image. Similar information can be obtained from the Lovastatin mask overlays: its outlier is also explained by a denser cell field.

Figure 16 also shows that DMSO fields from different plates are close in cell count, with approximately 250–300 objects per field and a median around 280. This is a high and relatively stable control. It suggests that the plates can be compared with each other, because their controls are similar.

Regarding the similarity between plates, we clearly see clusters by compound. Etoposide and AZ258 are especially compact, and they also have the lowest number of objects. This could be an important sign of drug effect if the effect is truly biological. It could be explained by reduced cell survival under drug treatment. However, this requires consultation with a biologist and validation on a larger sample. Our small dataset does not allow us to make such strong conclusions. We can only state that, in this work, the number of cells is visually consistent with the Cellpose mask overlays.

3.14 Morphometry collection

We confirmed that the segmentation can be considered valid. We can now extract data from the masks. The `skimage.measure.regionprops` method computes dozens of geometric and intensity-based characteristics for each cell in the mask. Table 7 lists the 11 most interpretable features that we will use in this work.

Before working with these features, however, incomplete cells must be removed so that they do not distort the data. Labeled objects that touch the image border are removed using the `clear_border` method from skimage.

Next, we filter out artefacts: objects that are either too small or too large. Small objects may correspond to cell fragments or imaging artefacts, while large objects may represent merged cells that were segmented incorrectly. To avoid removing biologically relevant cells, we estimate the threshold values from the expected diameters returned by Cellpose, which correspond to areas from 1,452 px to 7,284 px (area $\approx \pi \cdot \left(\frac{\text{diameter}}{2}\right)^2$).

Table 7: Morphometry features

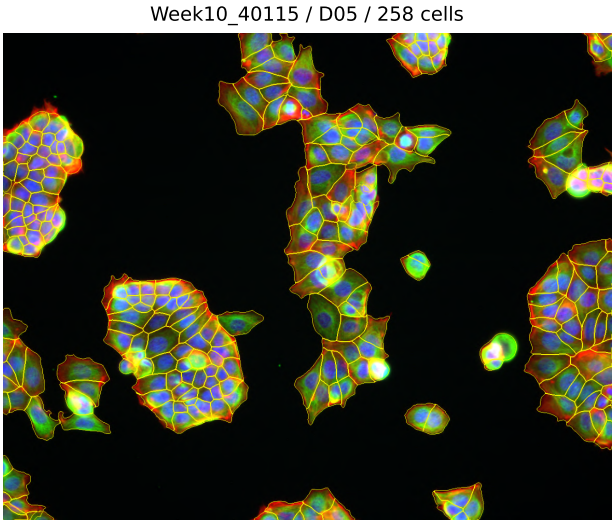
Feature	Meaning
`area`	Cell area in pixels
`perimeter`	Length of the cell boundary
`eccentricity`	How elongated the cell is: 0 to a circle, and 1 to a thread-like shape
`solidity`	`cell area / area of its convex contour`; how complex the contour is
`extent`	`cell area / area of the bounding box`
`equivalent_diameter`	Diameter of a circle with the same area
`major_axis_length`	Length of major axis of ellipse fitted
`minor_axis_length`	Length of minor axis of ellipse fitted
`aspect_ratio`	`major_axis_length / minor_axis_length`; a measure of elongation
`mean_actin`	Average brightness of the actin channel
`mean_tubulin`	Average brightness of the tubulin channel

Table 8: Area distribution quantiles

	0.01	0.05	0.50	0.95	0.99
DMSO	2.09	1328.65	3539.0	7011.10	8663.34
Lovastatin	429.15	923.90	2280.5	4130.85	6342.13
AZ-U	295.85	588.00	1452.0	3238.30	4059.16
Etoposide	1.00	2218.85	6009.5	12773.75	17691.27
AZ258	1.00	2331.40	9057.0	15312.85	17638.24

We also use the area distribution by quantiles as a reference point (Table 8). We can see that for three compounds the area goes down to 1–2 pixels or less, while the largest objects reach 17,000 pixels or more.

We choose wider conservative limits: objects with `area < 300` px and objects with `area > 25000` px are removed. This keeps the tails of the distributions in the violin plots while excluding obvious artefacts.



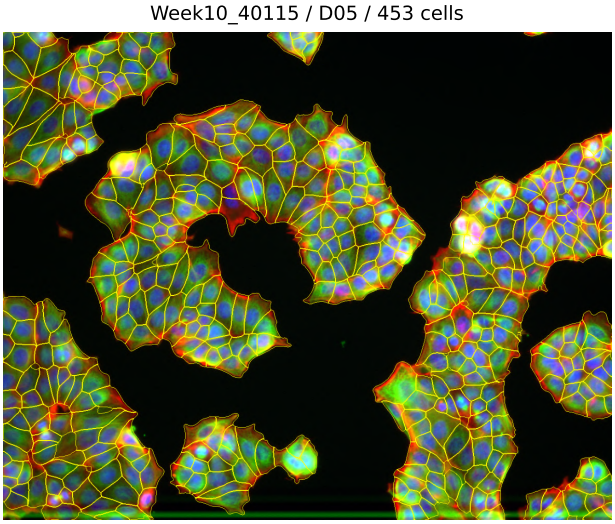


Figure 17: Outliers AZ-U

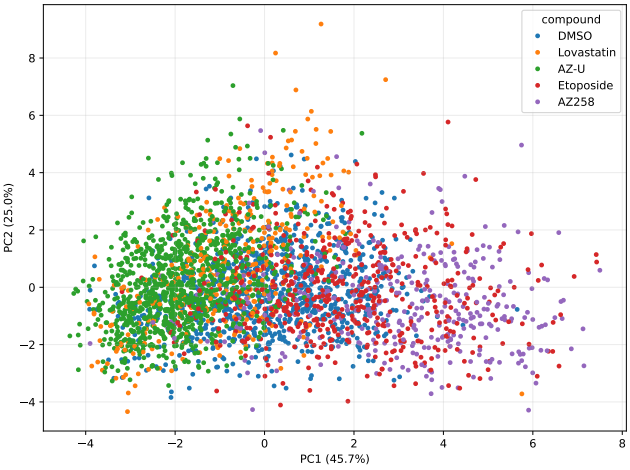


Figure 18: PC1/PC2

4 Results

4.1 Exploratory data analysis

We now have a cell-features table. This allows us to address the main question of the project: **how cell morphology differs under different drug treatments.**

To do this, we build violin plots for the selected features, apply PCA dimensionality reduction to test whether the treatments are separable by class, build pairplots to check whether the treatments separate by combinations of features, and summarize the z-normalized mean feature values by compound in a heatmap.

The feature violin plots in Figure 20 show that there are differences between the treatments. The main separation is observed in size-related features: `area`, `perimeter`, `equivalent_diameter`, `major_axis_length`, and `minor_axis_length`, which are expected to be strongly correlated. AZ-U shows the smallest phenotype, Lovastatin is closest to it, whereas AZ258 and Etoposide have the largest cells, with DMSO lying in between.

The separation by shape is weaker. `eccentricity` and `aspect_ratio` strongly overlap between compounds, meaning that elongation alone does not separate the classes well. `solidity` is more informative: Lovastatin has a noticeably wider lower tail, which may indicate more irregular or complex masks, or differences in cell shape that is actually true and seen on the images.

The intensity features, `mean_actin` and `mean_tubulin`, also show shifts between groups, but these differences are less clean than those observed for size-related features. `mean_tubulin` appears more useful than `mean_actin`, although the overlap remains substantial.

The z-heatmap in Figure 21 confirms what is visible in the violin plots. Here, size-related features also separate the groups most strongly. AZ258 is sharply above average in `area`, `perimeter`, `equivalent_diameter`, `major_axis_length`, and `minor_axis_length`; Etoposide is also above average, but less strongly. AZ-U, in contrast, is below average for almost all size features. DMSO is close to zero.

Lovastatin is interesting not so much in terms of size, but in terms of shape: it has strongly negative `solidity` (-0.98) and reduced `extent`. This is consistent with its more irregular, less compact cells with complex shapes.

There is no separation in `eccentricity` and `aspect_ratio`: the values are close to zero. This means that elongation is not the main driver of the differences.

In terms of intensity, DMSO and AZ-U are higher in `mean_tubulin`, Lovastatin is slightly higher in `mean_actin`, and AZ258 is lower in `mean_actin`.

Interestingly, PCA in Figure 18 shows strong overlap between the groups, whereas the pairplots in Figure 19 separate the data visually more clearly than the PC1/PC2 projection. The feature pairs show that some classes differ in cell size, shape, and tubulin intensity, although the overlap is still substantial. PCA is likely less suitable for this specific comparison because it is not designed to separate classes; it only finds the axes of maximum variance, rather than the best boundaries between compounds.

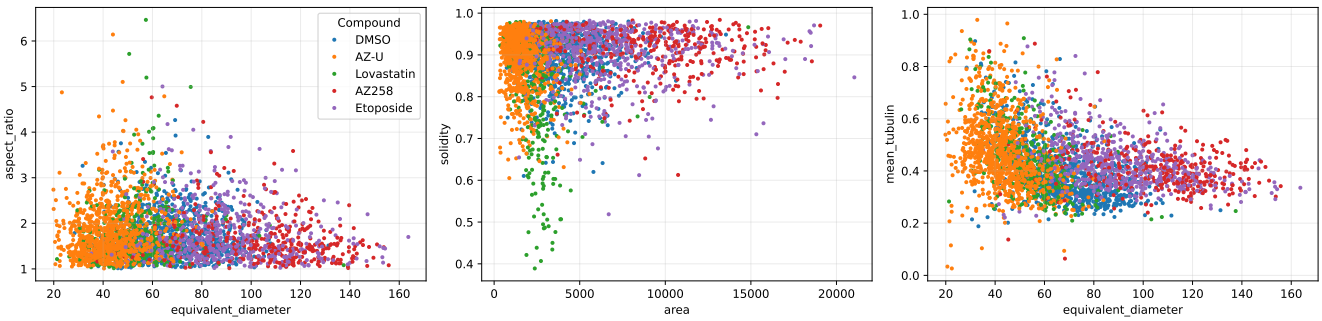


Figure 19: Morphometry PCA informative pairs

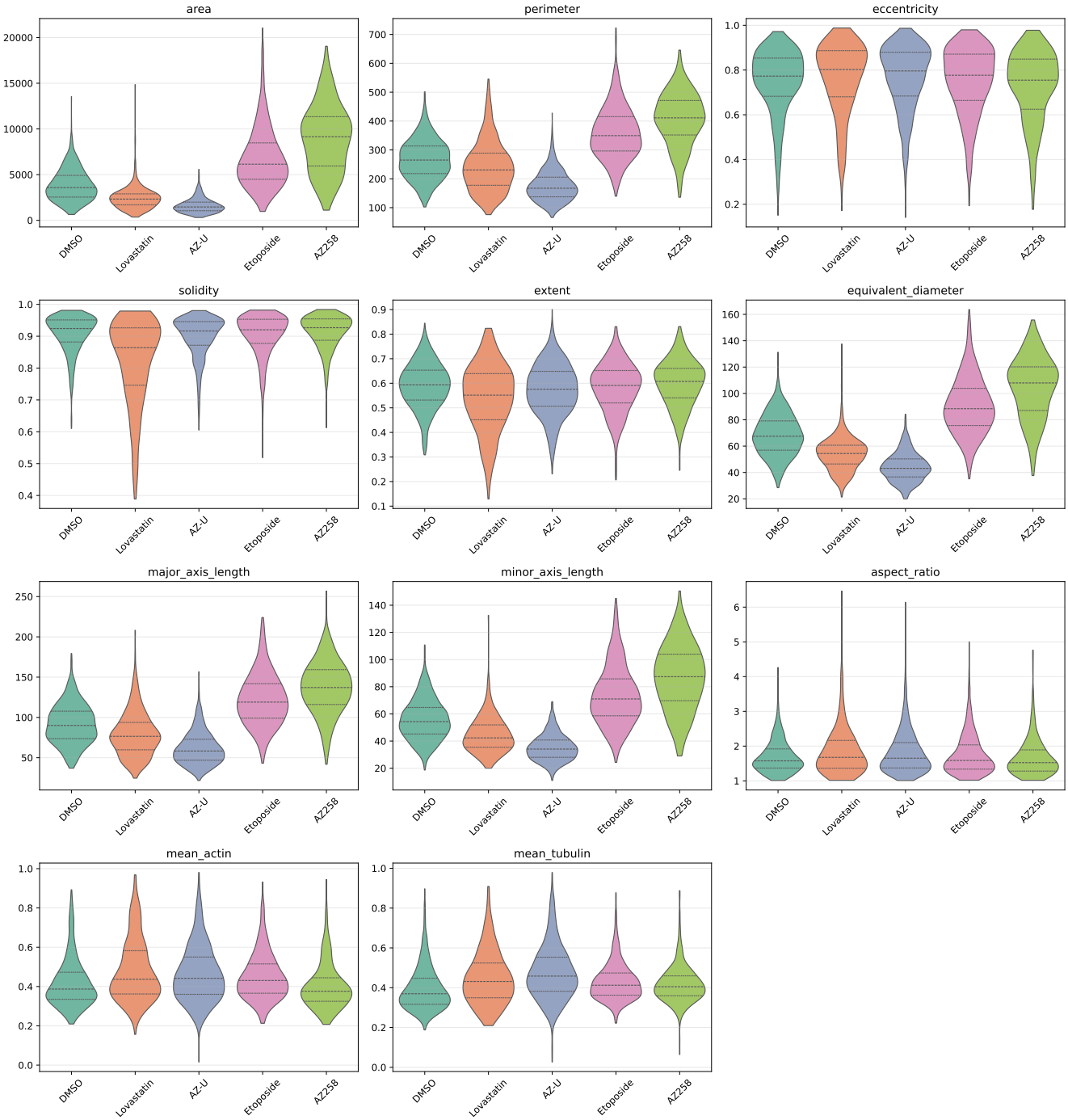


Figure 20: Morphometry violins

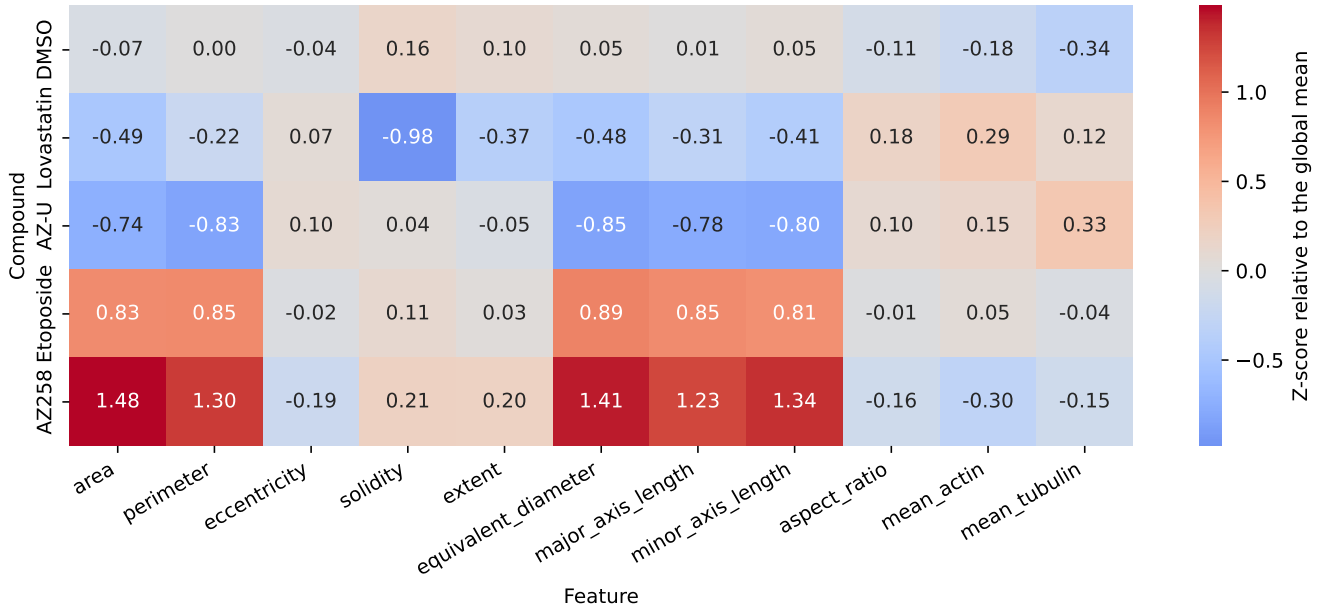


Figure 21: Morphometry heatmap



Figure 22: Confusion matrices

4.2 Classifiers

We build simple classifiers using the extracted features to test whether the classes of compounds used to treat the cells can be distinguished based on morphology.

First, it is important to split the data into training and test sets to avoid data leakage. We split the data by `field_id`, stratified by `compound`. This means that cells from the same field of view cannot appear in both the training and test sets at the same time. We have 4 fields of view per compound, so for each compound, 3 fields are used for training and 1 field is used for testing. If more data were available, we could also stratify by well, since wells may not be fully independent.

Table 9: Train-test split by field

	compound	train_fields	test_fields	train_cells	test_cells
0	DMSO	3	1	701	198
1	Lovastatin	3	1	269	68
2	AZ-U	3	1	812	234
3	Etoposide	3	1	404	125
4	AZ258	3	1	288	80

We train four classifiers: Logistic Regression as a simple linear baseline model, Random Forest as a nonlinear baseline model, SVM as a nonlinear margin-based classifier, and Decision Tree as an interpretable model.

Separately, we also train drug-only models without `DMSO`: an SVM and a Random Forest. This allows us to test how well the models distinguish only active treatments without the control class.

For models that are sensitive to feature scale, such as Logistic Regression and SVM, we standardize the features using `StandardScaler()`. We use `class_weight="balanced"` to compensate for class imbalance, so that errors on rarer classes are penalized more strongly.

By default, **Logistic Regression** uses `penalty=L2`: this penalizes overly large model coefficients and helps reduce overfitting. The model also uses standard moderate regularization with `C=1.0` by default: a smaller `C` means stronger regularization, while a larger `C` means weaker regularization.

The **Random Forest** classifier is an ensemble of many

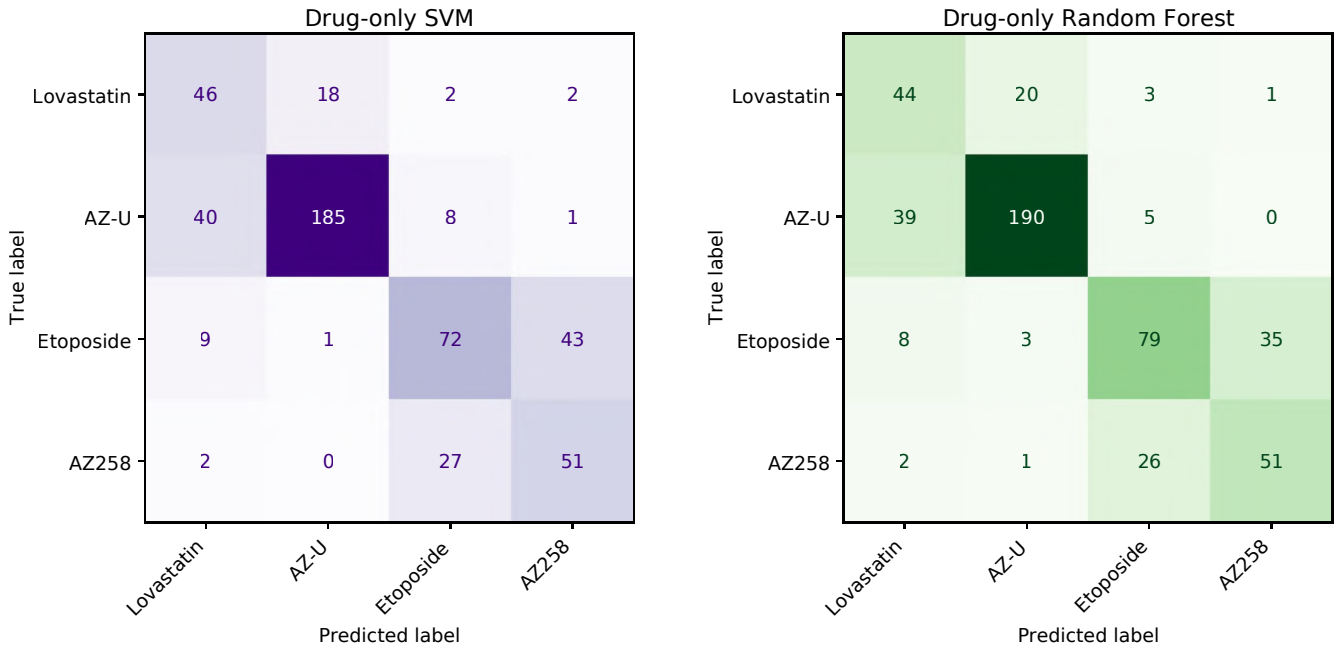


Figure 23: Drug-only confusion matrices

decision trees. Each tree tries to classify a cell based on its morphometric features, and the final prediction is made by voting across the trees. This classifier can capture nonlinear relationships and combinations of features. For example, if `area` is large, `mean_tubulin` is low, and `solidity` is high, the cell may be more likely to belong to `AZ258`.

`n_estimators=300` defines the number of trees in the forest. This value was selected experimentally. The same applies to `max_depth=8`. Tree depth means the number of consecutive split decisions from the root of the tree to a final leaf. In our case, values from 4 to 8 were treated as a reasonable experimental range: with 5 classes, depths below 4 are too shallow to separate the data in principle, while depths above 8 can increase overfitting.

Random Forest uses bootstrap samples. A bootstrap sample is a training subset created by sampling from the original training data with replacement. This means that each tree is trained on a slightly different version of the training set: some cells may appear more than once, while others may be left out. This makes the trees less identical and improves the stability of the ensemble.

A leaf is the final node of a tree where the prediction is made. The smaller `min_samples_leaf` is, the narrower and more specific the rules can become, which increases the risk of overfitting. The larger this parameter is, the less detailed the tree becomes, but the better it may generalize. Thus, `min_samples_leaf=10` prevents the trees from creating rules based on individual cells.

We apply **Support Vector Machine** with an RBF kernel, which allows the model to build nonlinear boundaries between classes. The idea is to maximize the margin between the decision boundary and the closest training points, called support vectors. The RBF kernel allows the model to separate classes using complex curved boundaries.

The regularization parameter `C=1.0` controls the trade-off between a simpler decision boundary and fewer errors on the training set. A smaller `C` means stronger regularization, producing a simpler and more stable boundary. A larger `C` means weaker regularization: the model tries harder to classify the training points correctly, which increases the risk of overfitting.

On the other hand, the `Gamma` parameter controls how local the influence of individual training points is. A small `gamma` gives a smoother and broader decision boundary; a large `gamma` gives a more complex boundary that adapts more strongly to individual points. We use `"scale"`, meaning that sklearn calculates `gamma` automatically.

Across all classifiers, the confusion matrices (Figure 22, Figure 23) show that the models are not guessing the classes randomly: most values lie on the diagonal, meaning that many cells are classified correctly. `AZ-U` is separated best.

The main problem is that not all compounds have a unique morphometric profile based on these 11 features. `DMSO` is often confused with `Etoposide` and `Lovastatin`; `Lovastatin` is noticeably confused with `AZ-U`; and the strongest pair of errors is `Etoposide` versus `AZ258`.

This pattern is repeated across all classifiers, which suggests that the limitation is more likely related to the data or the feature set rather than to the models themselves. The morphometric features provide a useful signal, but it is not sufficient to fully separate all treatment phenotypes at the single-cell level.

In the drug-only models without `DMSO`, the models still separate `AZ-U` well. `Etoposide` and `AZ258` remain a closely related pair. The main new observation is that `Lovastatin` is strongly confused with `AZ-U`. Thus, removing the control class does not solve the problem of

similar morphometric profiles among active treatments; instead, it makes this problem more visible.

The model performance metrics in Table 10 and Table 11 confirm that the classes overlap, although a useful signal is still present. Random Forest performed best. Macro F1 is lower than weighted F1 for almost all models. This means that the models recognise some individual classes less well, while the overall accuracy is largely supported by the easier and more numerous classes, AZ-U and DMSO. The F1-scores show that Etoposide and AZ258 are weakly separated, while Lovastatin is the weakest class overall.

Table 10: Overall classifier performance

Model	Accuracy	Macro avg F1	Weighted avg F1
Drug-only RF	0.72	0.67	0.72
Drug-only SVM	0.70	0.65	0.71
Random Forest	0.67	0.63	0.67
SVM	0.65	0.61	0.66
Log Reg	0.63	0.59	0.64

Table 11: F1-scores

Model	DMSO	Lovastatin	AZ-U	Etoposide	AZ258
Drug-only RF	–	0.55	0.85	0.66	0.61
Drug-only SVM	–	0.56	0.84	0.62	0.58
Random Forest	0.70	0.52	0.79	0.55	0.58
SVM	0.69	0.49	0.81	0.52	0.56
Log Reg	0.67	0.47	0.78	0.50	0.52

5 Conclusion and perspectives

Overall, this project developed an image-analysis pipeline for drug-treated cancer cell microscopy images, including image preprocessing, Cellpose-based segmentation, extraction of single-cell morphometric features, exploratory analysis, dimensionality reduction and treatment classification. The results show that the extracted morphology features contain a useful treatment-related signal, although the overlap between several classes limits classification performance.

In future work, the analysis could be strengthened by including a larger number of plates, wells and drug compounds, which would make the comparison more robust and reduce the risk that the observed patterns are driven by a narrow subset of images.

It would also be valuable to integrate human-guided annotation or segmentation tools such as MedSAM, or similar services, and compare these annotations with the fully automatic masks. This would help assess segmentation quality more directly.

We also began to check (in Appendix) whether cell morphology differs across concentrations. We originally downloaded the subset so that each compound, except DMSO, had three concentrations. For the main analysis, we used the middle concentration. We then use all three concentrations and plot the number of segmented cells and morphological features as a function of concentration. This analysis could be explored in more depth in the future.

Code Availability

The code is available on Google Drive.

Use of AI

Artificial intelligence tools were used to support language polishing, wording refinement, and consistency of the written report.

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Appendix

Table 12: Short list of compounds, their MoA groups, and concentrations

	moa	compound	concentrations
4	Aurora kinase inhibitors	AZ258	[0.1, 0.3, 1.0]
5	Aurora kinase inhibitors	AZ841	[0.1, 0.3, 1.0]
6	Cholesterol-lowering	mevinolin/lovastatin	[1.5, 5.0, 15.0]
7	Cholesterol-lowering	simvastatin	[2.0, 6.0, 20.0]
11	DNA damage	etoposide	[1.0, 3.0, 10.0]
13	DNA replication	camptothecin	[0.003, 0.01, 0.03]
19	Epithelial	AZ-J	[1.0, 3.0, 10.0]
20	Epithelial	AZ-U	[1.0, 3.0, 10.0]
29	Microtubule stabilizers	docetaxel	[0.03, 0.1, 0.3]
30	Microtubule stabilizers	epothilone B	[0.1, 0.3, 1.0]
31	Microtubule stabilizers	taxol	[0.3, 1.0, 3.0]
37	Protein synthesis	cyclohexamide	[5.0, 15.0, 50.0]
38	Protein synthesis	emetine	[0.1, 0.3, 1.0]

Table 13: Precision and recall

Model	DMSO P	DMSO R	Lov. P	Lov. R	AZ-U P	AZ-U R	Eto. P	Eto. R	AZ258 P	AZ258 R
Drug-only RF	–	–	0.47	0.65	0.89	0.81	0.70	0.63	0.59	0.64
Drug-only SVM	–	–	0.47	0.68	0.91	0.79	0.66	0.58	0.53	0.64
Random Forest	0.73	0.67	0.46	0.60	0.86	0.74	0.53	0.57	0.53	0.64
SVM	0.78	0.62	0.39	0.66	0.88	0.74	0.51	0.53	0.49	0.65
Log Reg	0.75	0.61	0.38	0.62	0.86	0.72	0.49	0.52	0.45	0.60

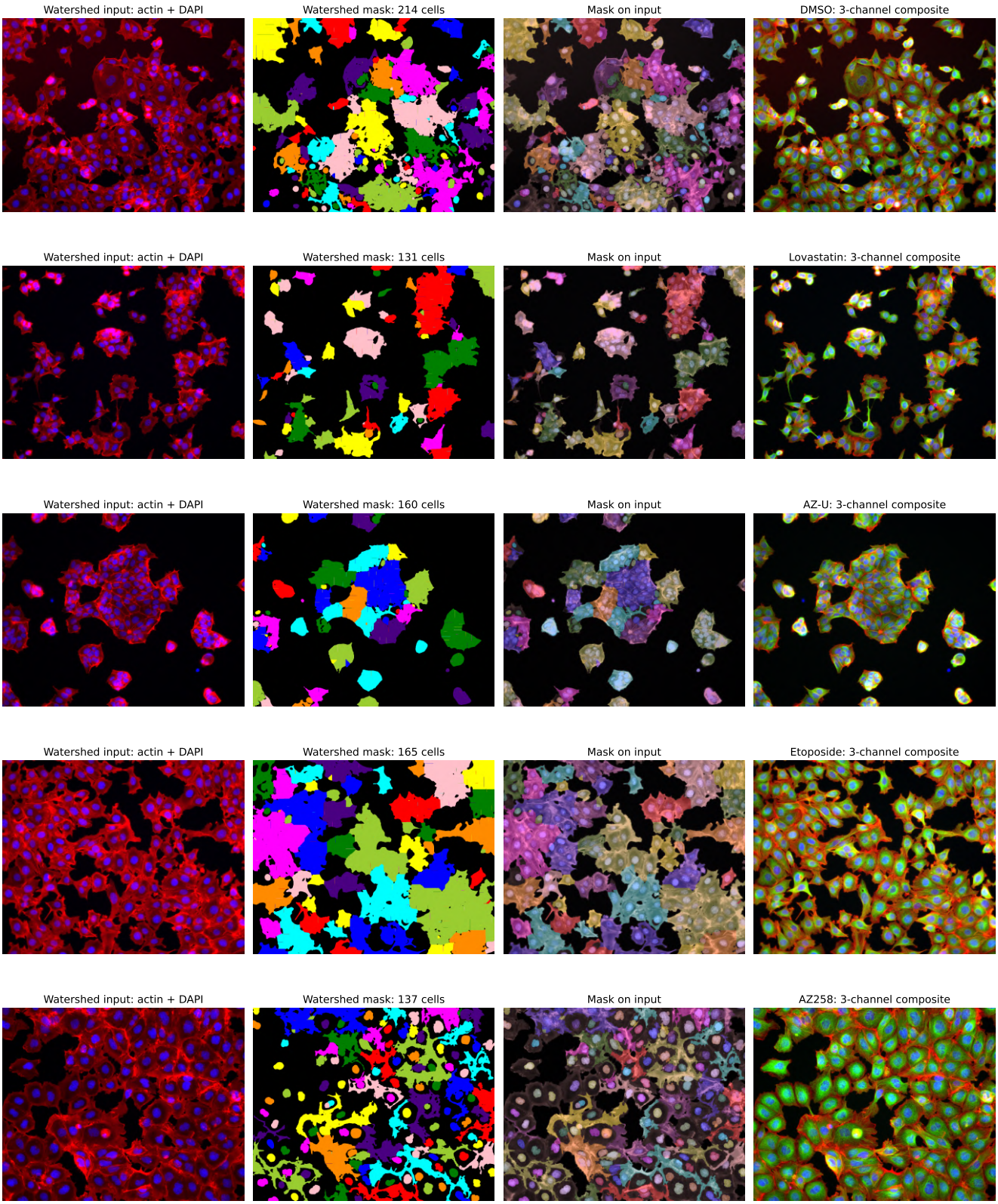


Figure 24: Naive watershed

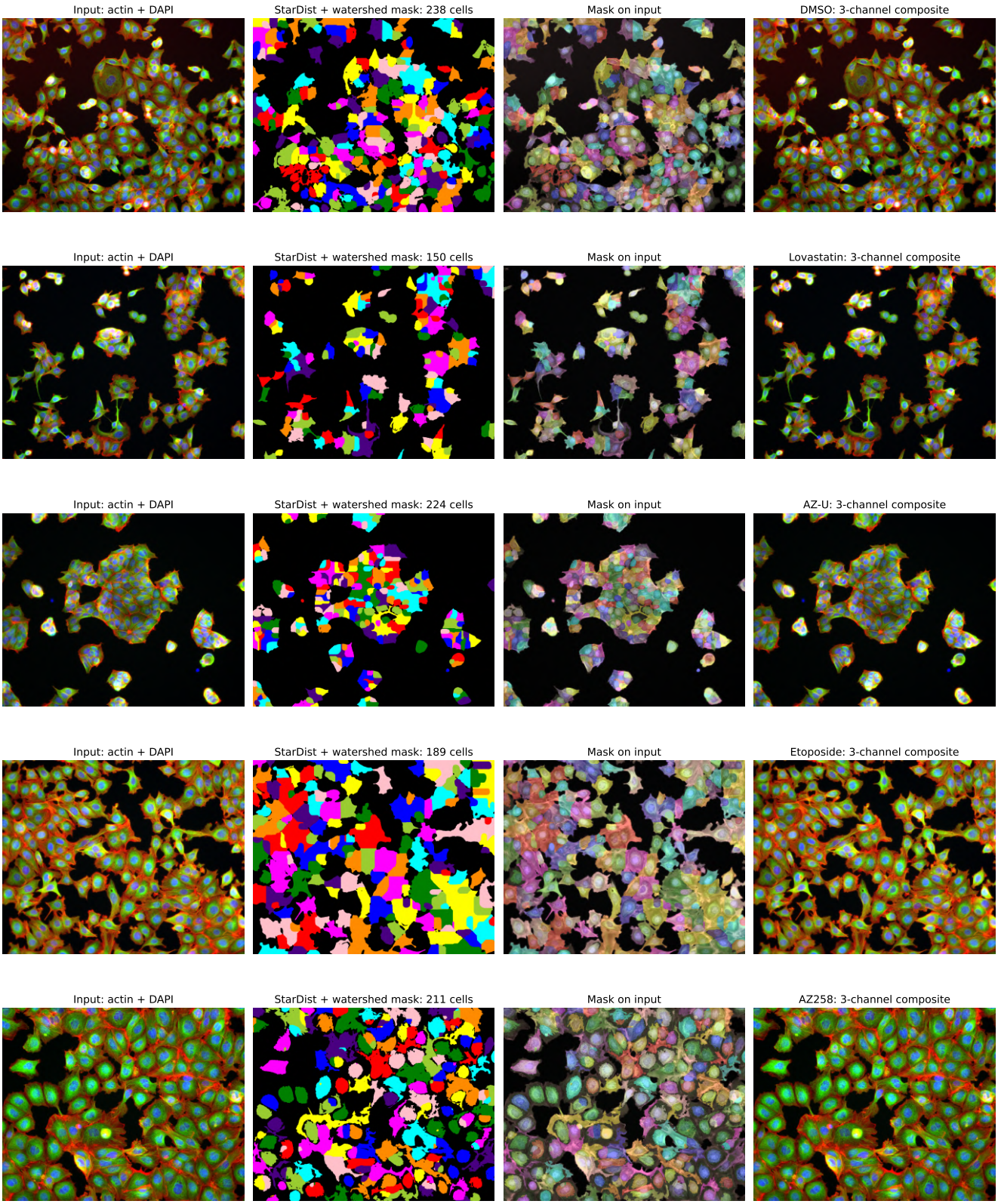


Figure 25: Starshed method (combined Watershed + StarDist)

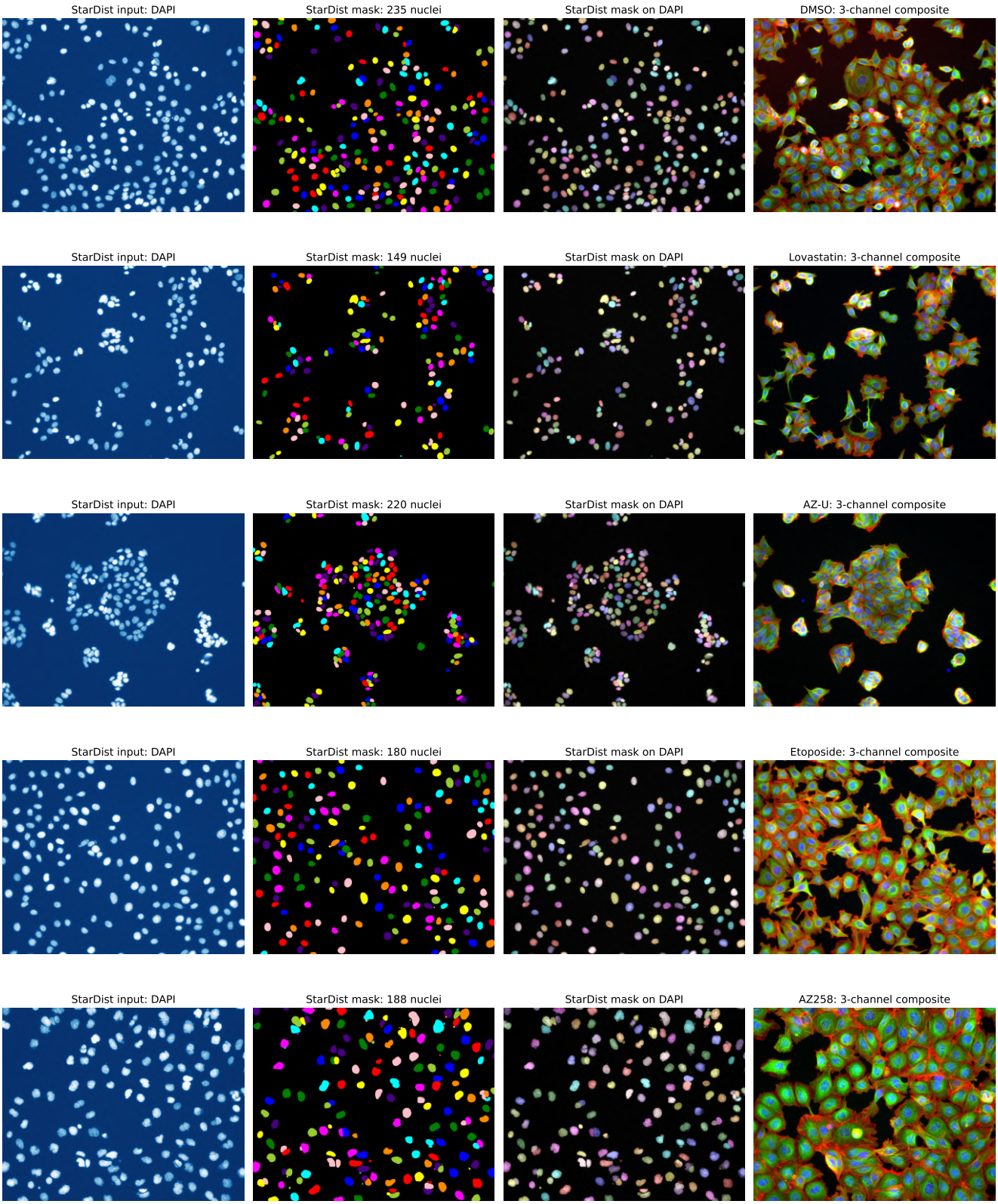


Figure 26: Stardist nuclei

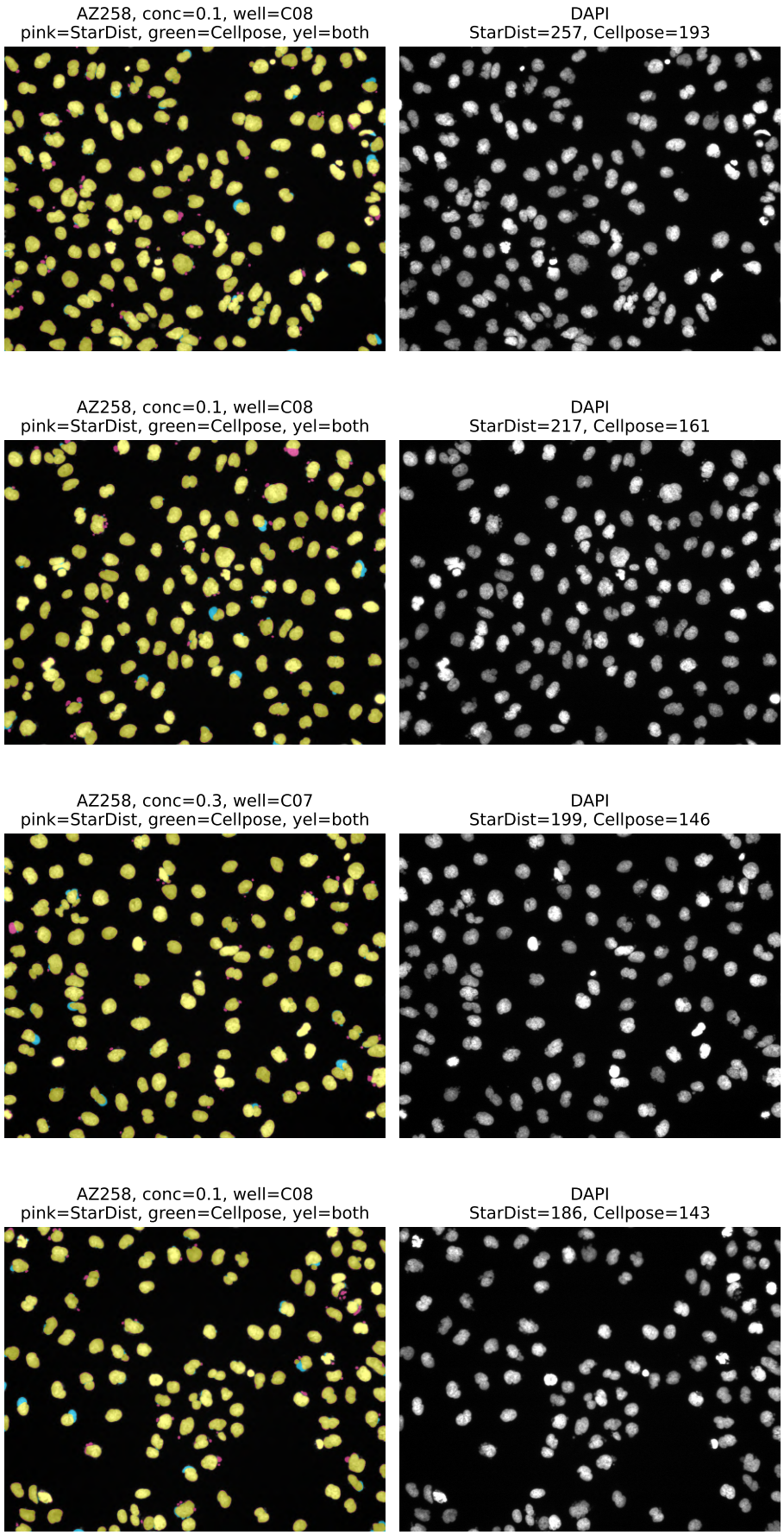


Figure 27: Nuclei segmentation comparison between Stardist and Cellpose for AZ258

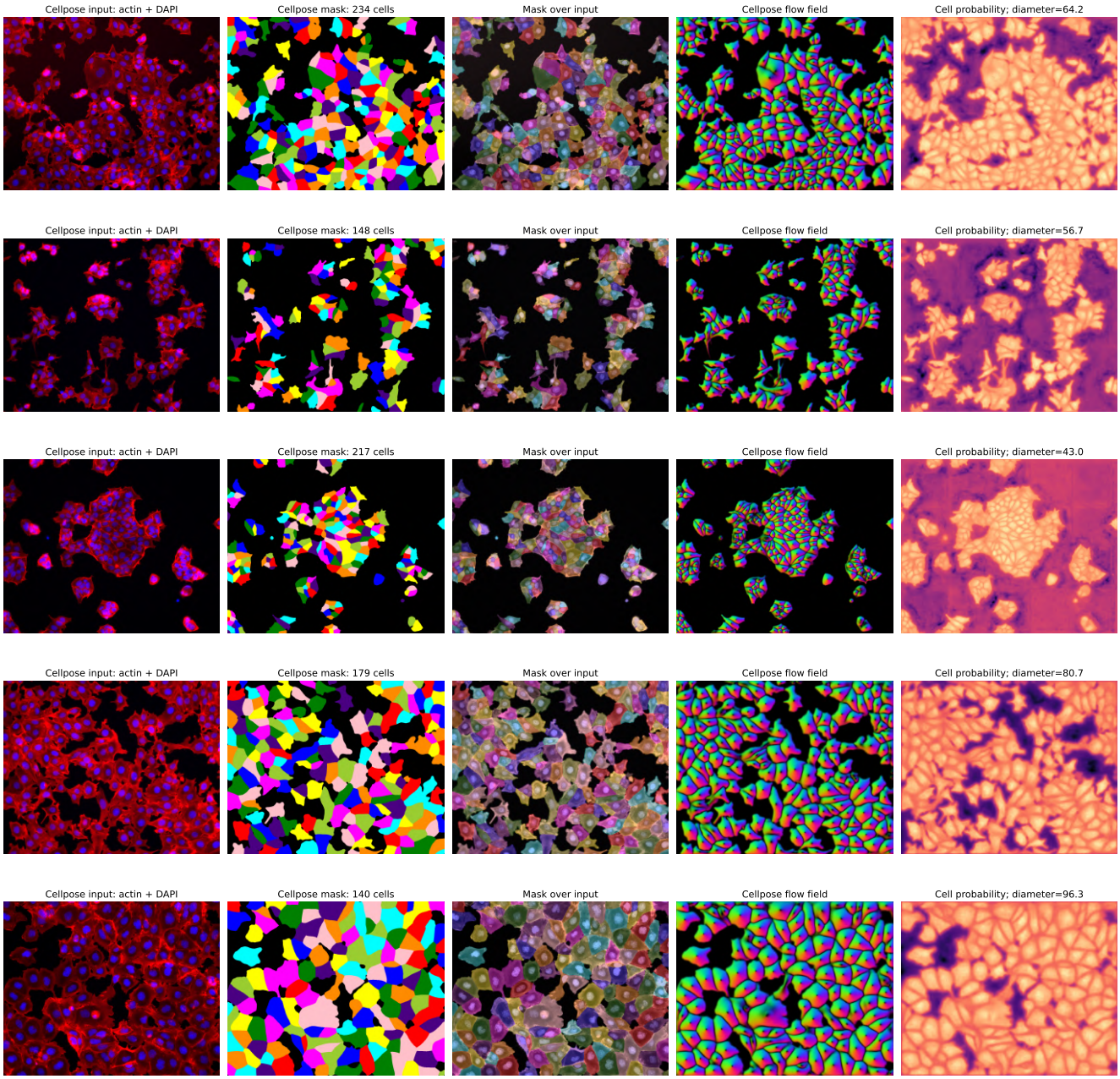
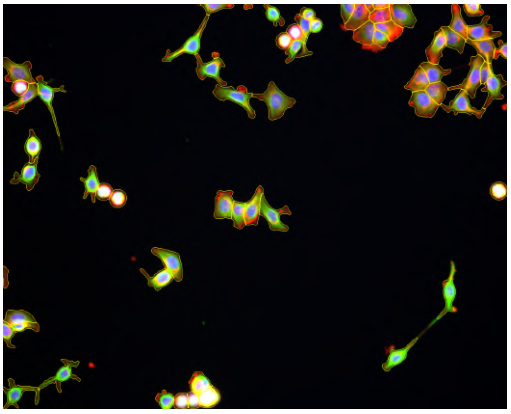


Figure 28: Cellpose flows and cell probability

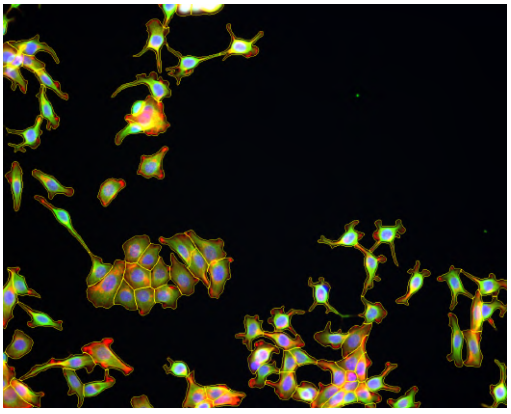
Table 14: PCA feature loadings

	PC1 (45.7%)	PC2 (25.0%)	PC3 (11.2%)
minor_axis_length	0.41	-0.20	0.12
major_axis_length	0.41	0.17	0.14
area	0.43	-0.08	0.16
perimeter	0.43	0.08	0.15
equivalent_diameter	0.44	-0.06	0.13
mean_actin	-0.23	-0.05	0.66
mean_tubulin	-0.22	-0.04	0.68
extent	-0.02	-0.49	-0.06
aspect_ratio	-0.01	0.52	0.02
solidity	-0.05	-0.40	-0.08
eccentricity	-0.01	0.50	-0.01

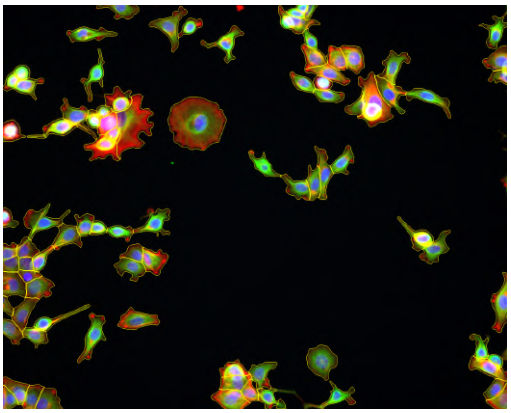
Week9_39221 / E05 / 67 cells



Week9_39221 / E05 / 92 cells



Week9_39221 / E05 / 92 cells



Week9_39221 / E05 / 163 cells

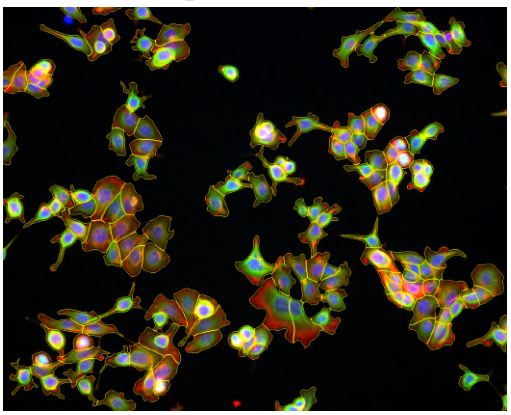


Figure 29: Outliers Lovastatin

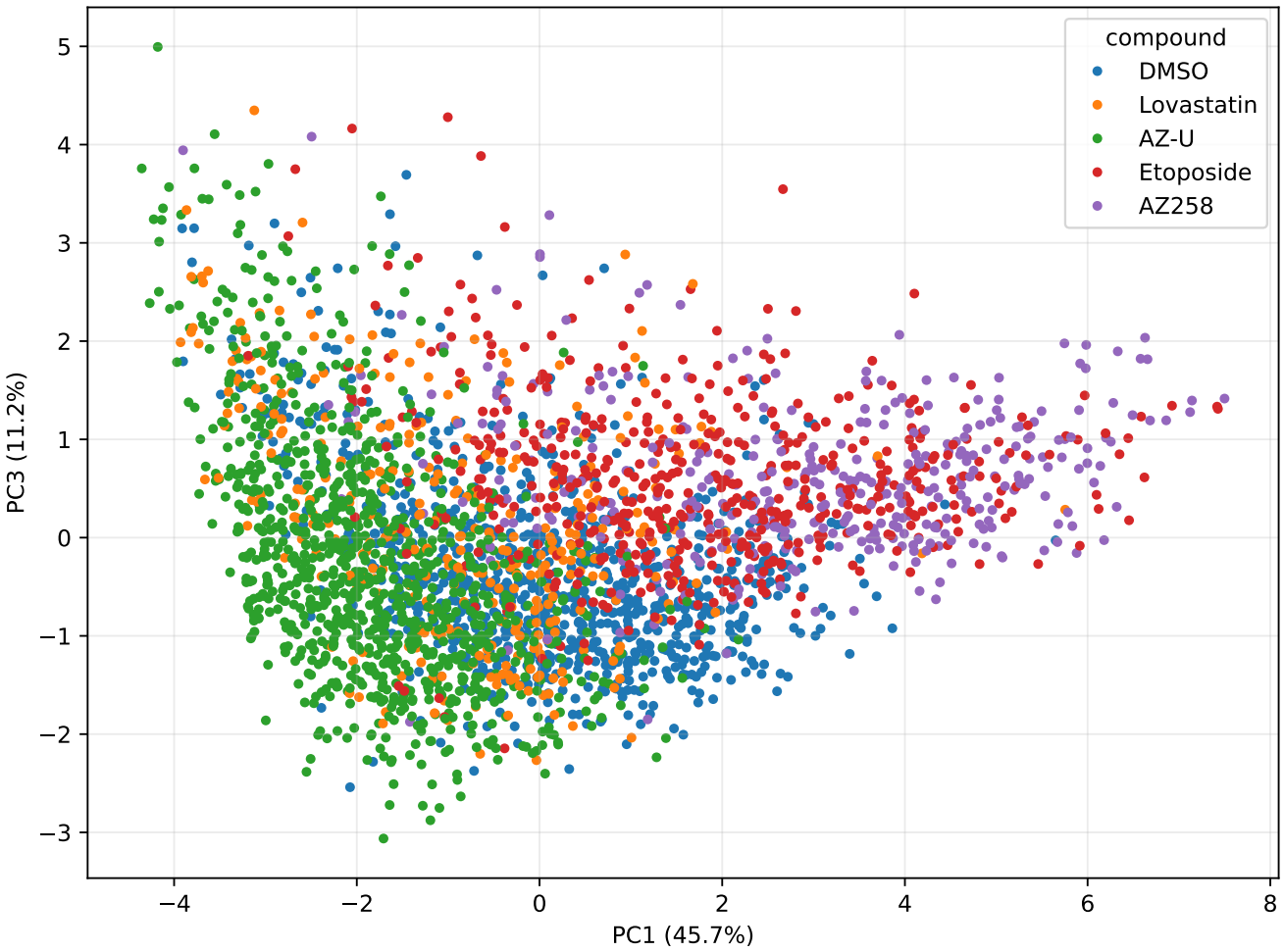


Figure 30: Principal Components 1 and 3

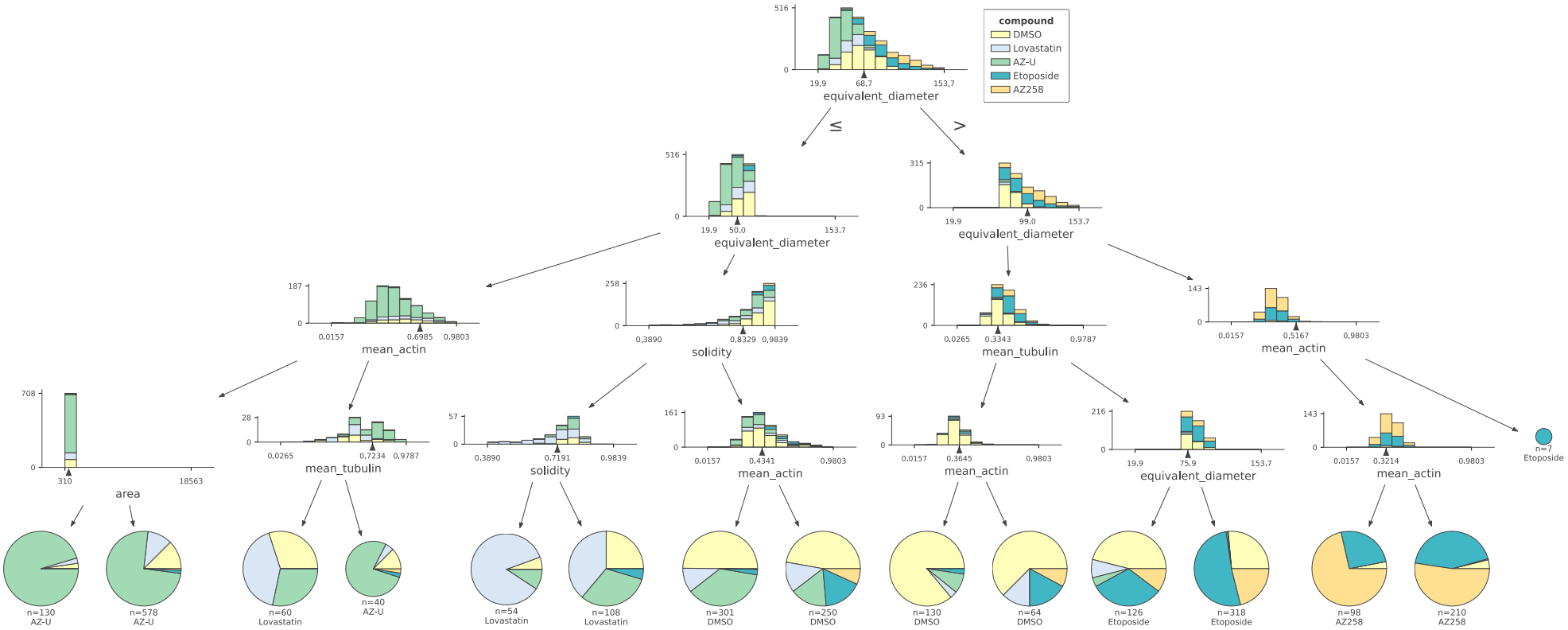


Figure 31: Decision tree dtreeviz

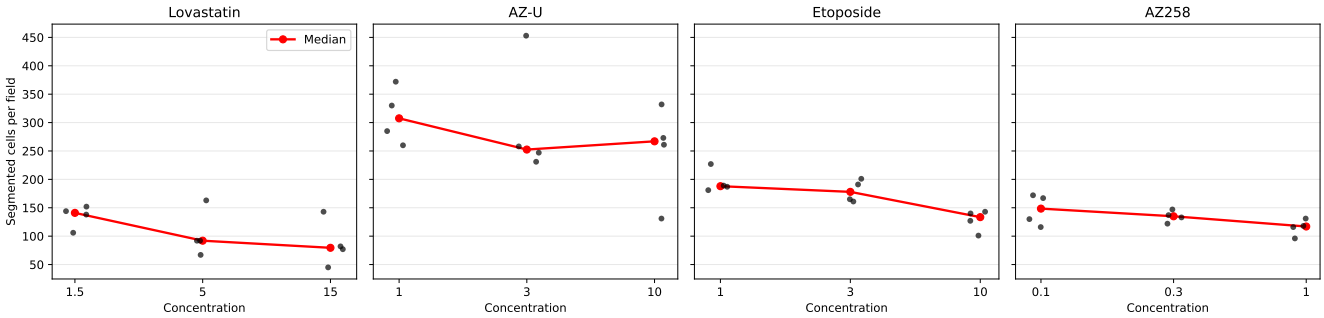


Figure 32: Dose response cell counts

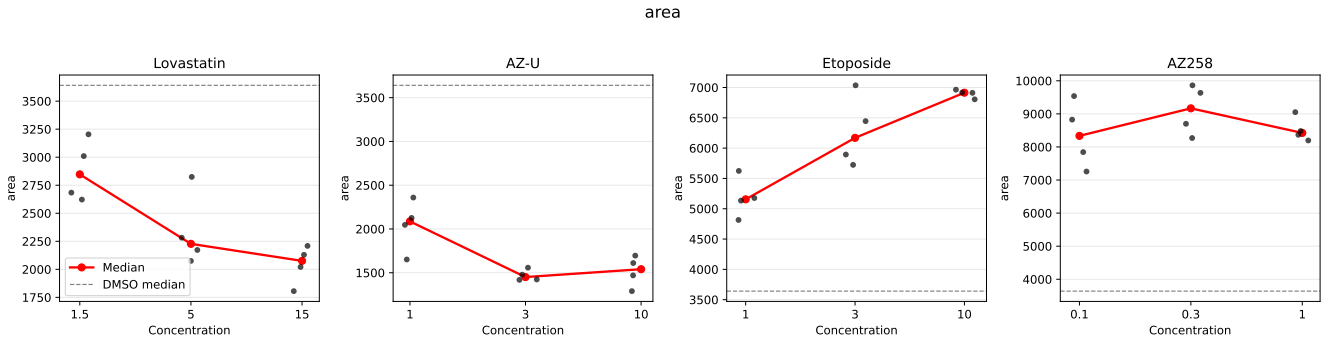


Figure 33: Dose response area

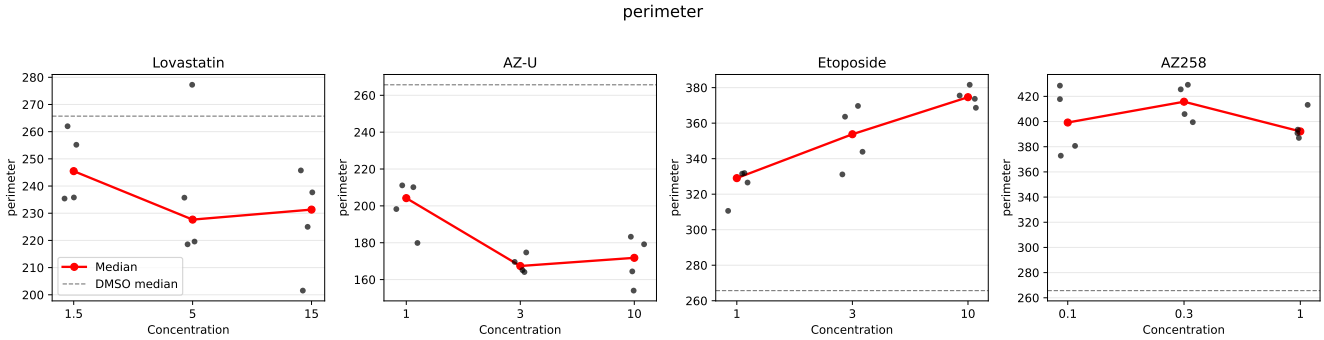


Figure 34: Dose response perimeter

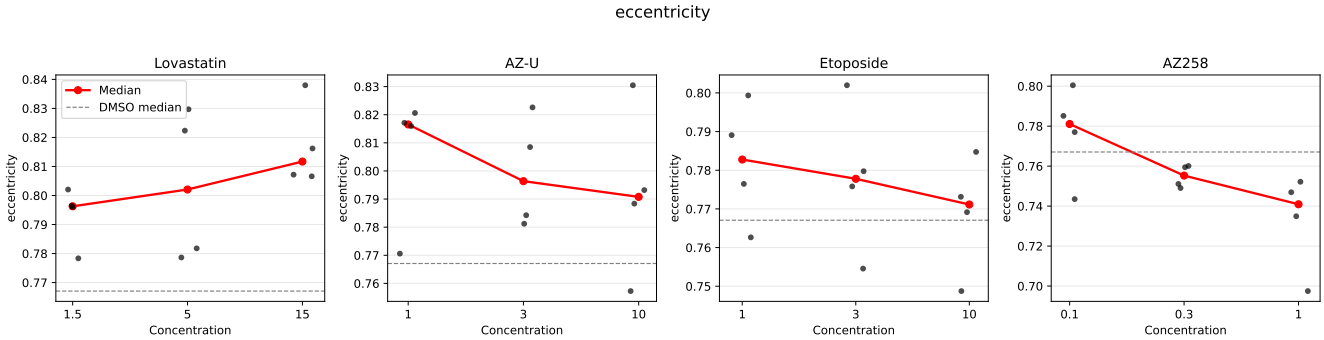


Figure 35: Dose response eccentricity

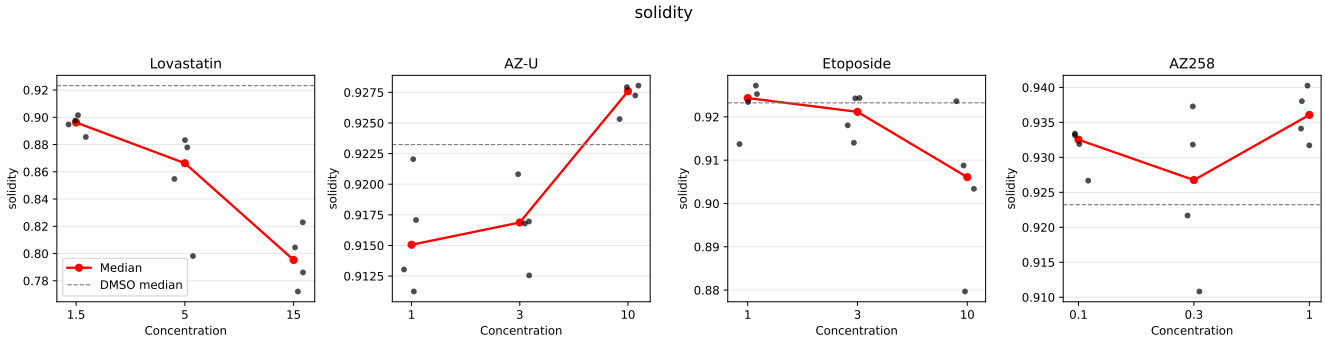


Figure 36: Dose response solidity

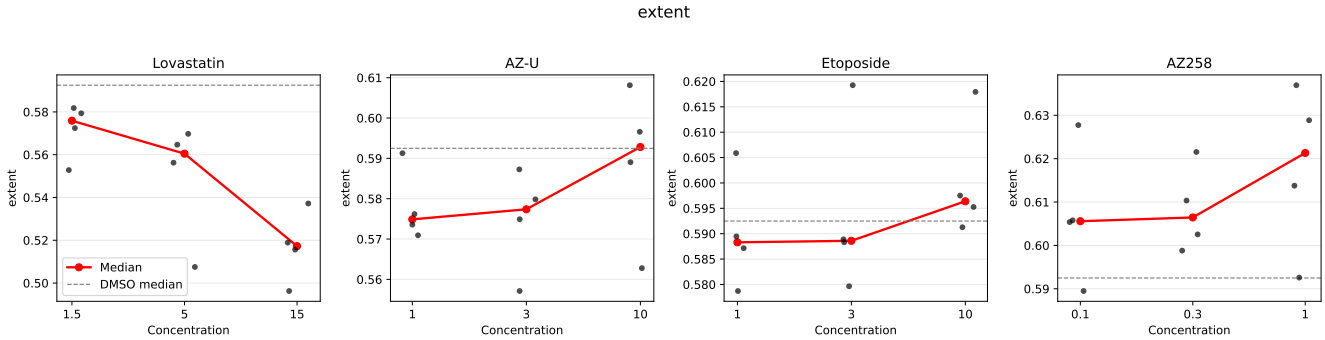


Figure 37: Dose response extent

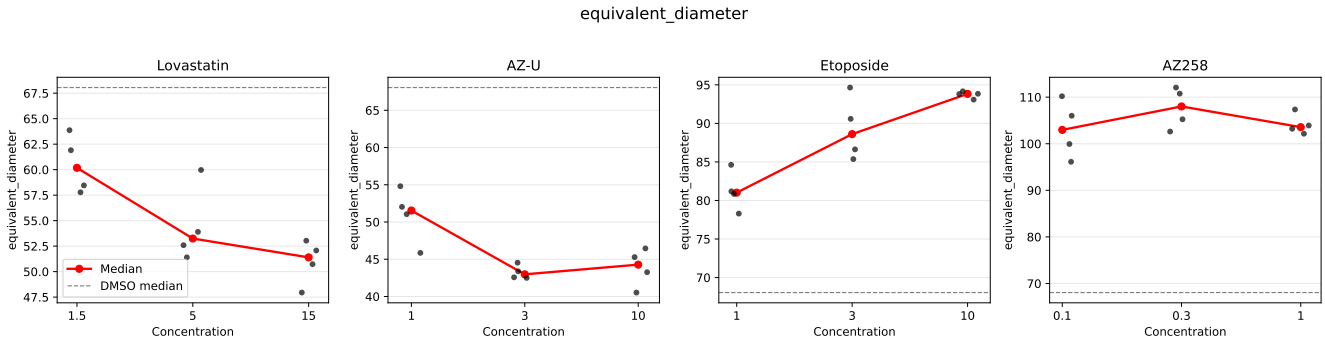


Figure 38: Dose response equivalent diameter

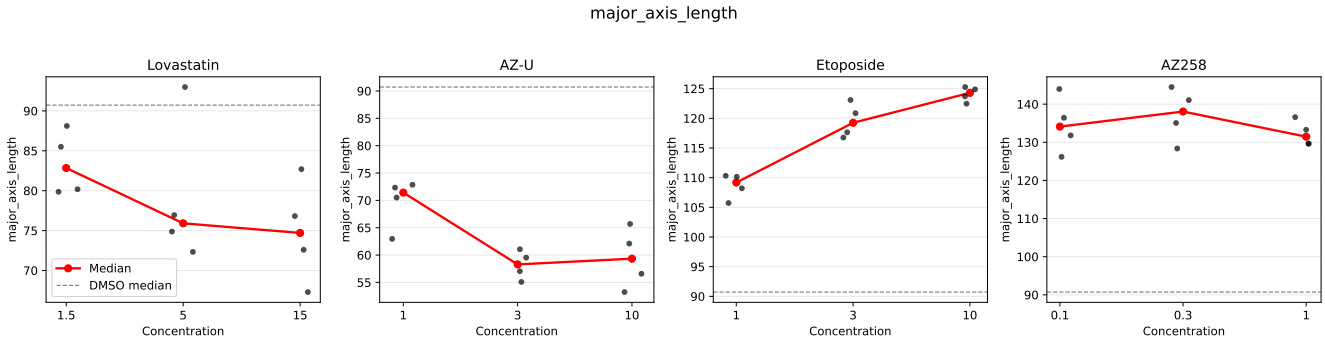


Figure 39: Dose response major axis length

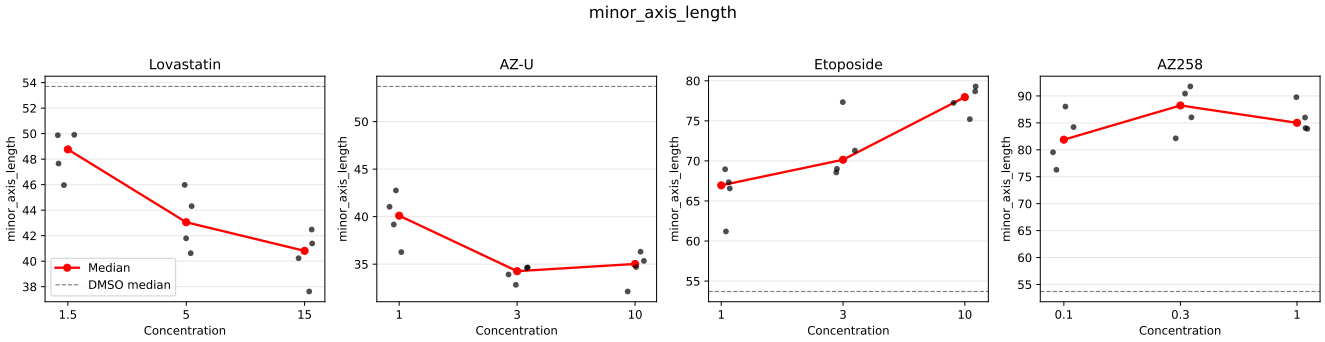


Figure 40: Dose response minor axis length

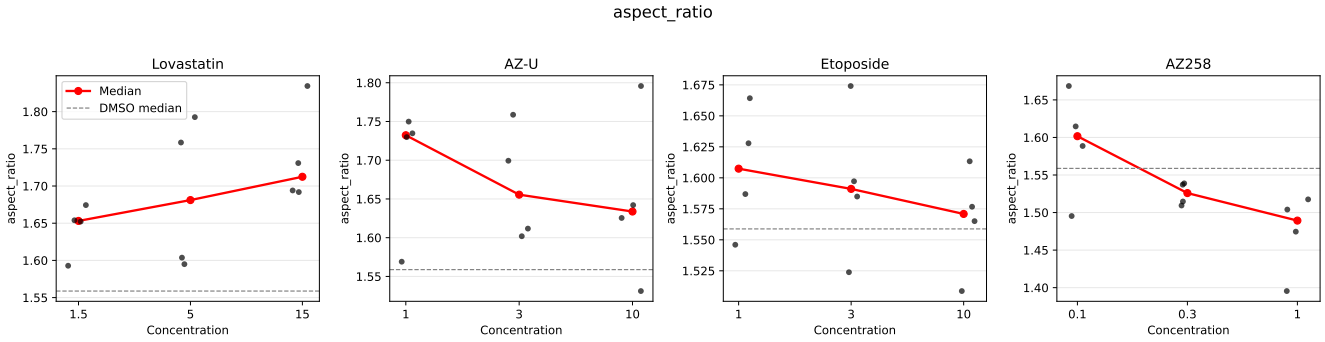


Figure 41: Dose response aspect ratio

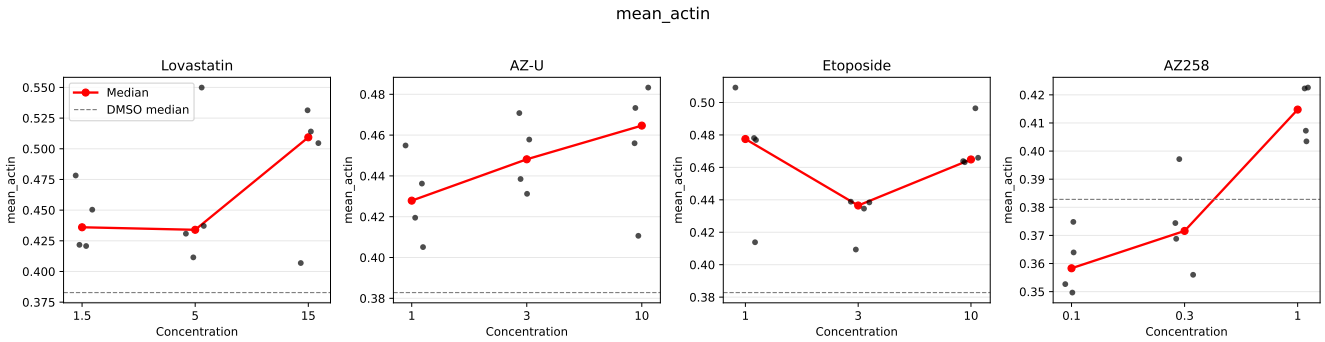


Figure 42: Dose response mean actin

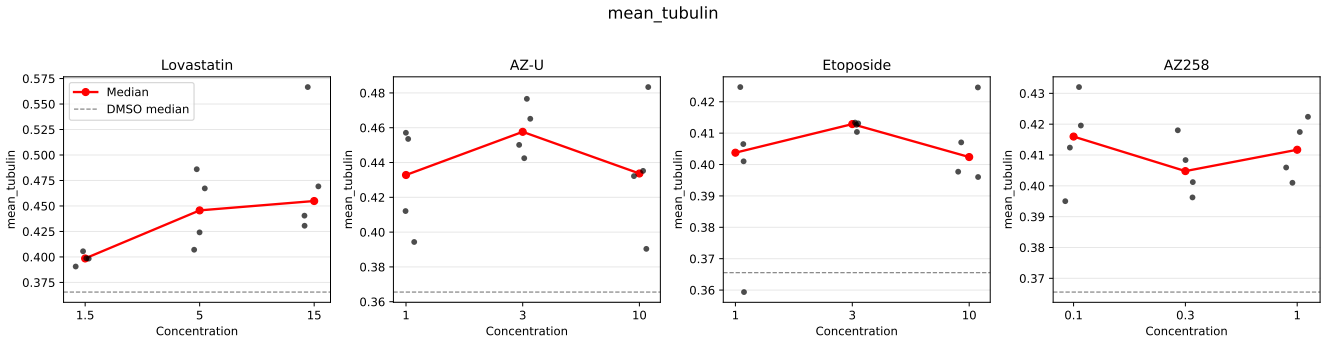


Figure 43: Dose response mean tubulin